Scotland versus the Rest:

Differential AI Exposure, Model Dependence, and the Scottish Budget

Tommy Ranyard

MSc Economics, University of Edinburgh

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Abstract

The block grant adjustment responds to the difference between the Scottish and rUK tax bases and not to the level of either, so any shock to the Scottish fiscal position must act through that difference. Artificial intelligence is of the like since it will not affect every occupation equally, and what is uneven across occupations is uneven across regions. Yet the available exposure measures are point estimates, reported without addressing how much of the score came from the model that produced it. This dissertation establishes which parts of such a measure institutions can rely on, and how to audit economic claims built on it. The solution for this paper's setting comes rests on a coincidence that differencing across regions removes the common component of any model-specific bias. Standardised or rank-based differentials remove any surviving affine bias that rescales the index but does not reorder occupations. Scotland sits two percentile points below the rUK in the occupational exposure distribution, 0.86 index points on a 0–100 scale. Re-scoring all tasks on two further models preserves the sign and the percentile magnitude but not the index-point figure. The gap implies a ten-year TFP differential of three basis points and a net-position sensitivity near £1.6 million a year, whose largest uncertainty is the choice of scoring model. Exposure is no material differentiator in next year's Scottish forecast, though its direction and persistence bear on a system reconciled *ex post*. Where magnitude does matter, the choice of model can shift the figures institutions lean on even more than the data do.

Keywords: artificial intelligence, occupational exposure, large language model scoring, model dependence, shift-share decomposition, Scotland.

JEL classification: C43, H68, H77, J24, O33, R23.

Contents

1 Introduction

Since the Scotland Acts of 2012 and 2016 devolved and partially devolved taxes now fund close to half of the Scottish Government's resource spending, with non-savings, non-dividend (NSND) income tax the largest part (Scottish Fiscal Commission 2026). Under the block grant adjustment (BGA), now the block grant is cut by foregone revenue, and that deduction indexed thereafter to the growth of the equivalent rUK tax base (Scottish Government and HM Treasury 2023). The Scottish budget therefore gains only insofar as Scottish income tax grows faster than its rUK comparator, such that the level of Scottish revenue is immaterial and only the differential matters — a pattern that underlies much of the theory and application of this paper. Forecast at £969 million for 2026–27, the Scottish Fiscal Commission (SFC) records this gap between devolved revenue and the block grant as the income tax net position. Thus, a relative deterioration in Scottish earnings shrinks the net position and since outturn is reconciled against forecast three years later, an error made now goes on constraining spending long afterwards (Scottish Government and HM Treasury 2023).

Artificial intelligence is a shock capable of impacting such a difference, both in the short and the long run. Any technology that raises productivity and alters labour demand unevenly across occupations will then also reshape earnings unevenly across the economy, with any resulting divergence between the Scottish and rUK tax bases surfacing in the aforementioned net position (Scottish Parliament 2016). There is good reason to expect such a divergence due to Scotland's distinctive occupational composition. The direction may culminate from the composition harbouring Scotland from displacement, or it may deny Scotland the gains that accrue to complemented work. This is furthered by a productivity gap that predates AI, with growth weak across Scotland, the UK, and much of Europe since the 2008 crisis meaning a worsened fiscal outlook (Scottish Fiscal Commission 2026). Whether AI narrows or widens that gap is a question that depends on how its gains are distributed across regions.

To assess this, the standard input is now an exposure score produced by a large language model, an approach Eloundou et al. (2024) established and institutional forecasters have since adopted, among them the Office for Budget Responsibility (2025) for the UK productivity outlook and the ILO for a global index (Gmyrek et al. 2023). Those scores are reported as point estimates with dependence on the scoring model merely conceded, yet recent evidence unsettles the premise that they are objective measurements at all. Yin et al. (2026) score one task set with four LLMs and find average exposure differs by more than a factor of three, with the labour-market figures built on those scores varying roughly 2.4-fold. This divergence is systematic not random, and so averaging does not remove it. What, then, can a fiscal exercise like this one take from a number that changes with the model that produced it? This dissertation addresses that question, and answers it for this fiscal application. No instrument for AI exposure yet matches the commuting-zone design of Acemoglu and Restrepo (2020), so what follows is measurement rather than causal identification. The solution rests on a coincidence: the BGA responds to the Scotland–rUK *difference* not the Scottish *level*, and the exposure index is dependable in differences but not in levels.

Three contributions follow, with the first being analytical. Proposition 1 shows that the regional differential in an exposure index removes the common component of any model-specific bias, retaining only the part co-varying with the cross-region employment-share gap, with Corollary 1 bounding that part by a Cauchy–Schwarz inequality. Corollary 2 covers the case where a model disagrees about the scale of the index while agreeing about order, and there a rank-based differential removes the bias in full. I evaluate whether the result holds by re-scoring the full task set on two further language models. Secondly, I contribute a compositional answer to the fiscal question, locating Scotland fifth of the twelve UK regions and nations, attributing about two-fifths of the headline gap to London. Thirdly, I quantify the impact of AI in fiscal units, carrying scoring uncertainty through.

Scotland's employment-weighted continuous exposure sits 0.86 index points below the rUK on the 0–100 scale, with a 95% interval of $[-1.07, -0.69]$, with most of the width coming from survey sampling. The scale-free reading translates to about two percentile points of the UK occupational exposure distribution. The sign survives two model re-scores, the raw gap moving to -0.53 under Claude Haiku and -0.43 under GPT-4o as absolute levels shift by up to 9.1 index points. Passed through the Acemoglu calibration, the differential implies a ten-year TFP differential of around three basis points, which Section 6.2 translates into a net-position sensitivity of about £1.6 million a year. Scoring noise moves this figure by 9%, survey sampling and the adoption scenario by around a two-fifths each, and the choice of scoring model by 50%.

The rest of the paper is organised as follows. Section 2 sets out the fiscal backdrop, the task-based mechanism, and the measurement literature, ultimately proposing the three research gaps I address. Section 3 describes the data. Section 4 constructs the exposure scores, states the region-invariance assumption, proves the cancellation results, and sets out the calibration and propagation design. Section 5 presents the differential, its anatomy, its robustness across thresholds and scoring models, and the exposure–wage gradient. Section 6 carries the differential into projected productivity and then into the income tax net position, and quantifies uncertainty in the resulting figure. Section 7 concludes. Figure 1 depicts the paper's procedure from start to finish.

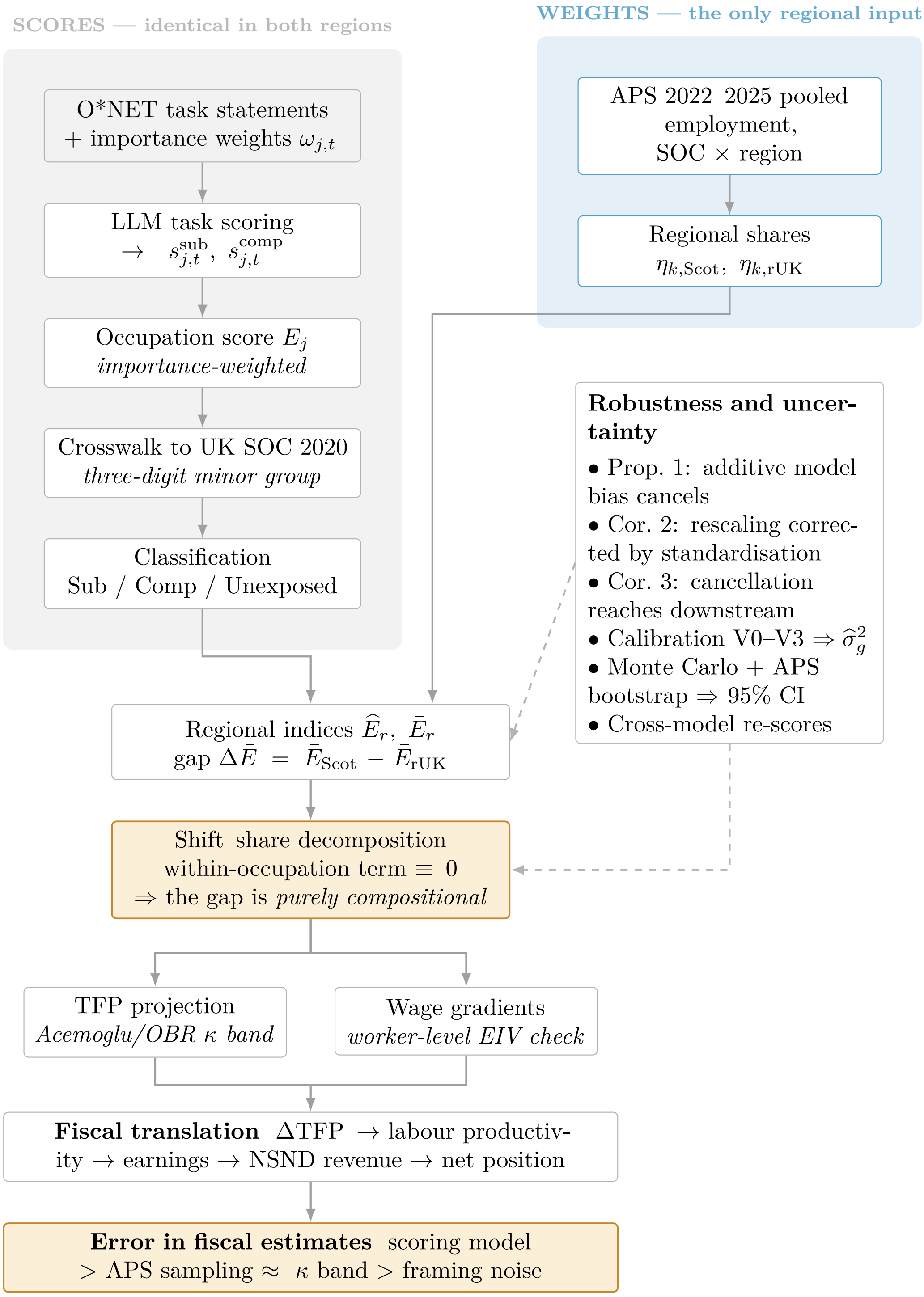

Figure 1: The empirical procedure, from task scores to the income tax net position.

2 Background and Related Literature

The task mechanism

The labour-market consequences of AI run through the task. Whether AI raises or lowers the demand for labour depends on which of the individual activities that make up the job it can complete, complement, or cannot do. In the Acemoglu and Restrepo (2019) task-based framework, production allocates a continuum of tasks between labour and capital, with technology acting through two opposing channels. While a displacement effect means capital absorbs tasks workers performed, a reinstatement effect means new tasks restore a comparative advantage for labour. Importantly, productivity gains and labour demand can move apart. "So-so" automation that displaces workers while delivering only modest productivity gains, has been shown to depress the labour share even as output rises (Acemoglu and Restrepo 2020). It is, thus, not enough to say exposure to AI implies harm since a task AI can perform is displaced, whereas a task AI makes a worker faster at is complemented, with each carrying opposite signs for labour demand.

Which tasks a technology reaches has changed. Earlier computerisation acted on codifiable tasks, as emphasised in the routine-biased technical change literature (Autor et al. 2003; Goos and Manning 2007; Autor and Handel 2013), hollowing out middle-wage clerical and production work while sparing both manual and analytical non-routine activity. Newer advances in generative AI reaches the non-routine cognitive tasks the previous wave could not, so exposure no longer splits down the routine/non-routine divide. The updated fault line is consequently the substitution–complementarity margin, and the literature debates on which dominates. Autor (2024) reads AI optimistically, as a tool that could reinstate middle-skill judgement and widen access to expert tasks, whereas Acemoglu (2025) is more cautious, projecting an aggregate total-factor-productivity gain of under one per cent over a decade once only currently automatable tasks are counted. The mechanism AI will operate along is less disputed, but the sign and magnitude of the impact is an ongoing empirical point of interest.

Measuring exposure and the reliability problem

Turning the mechanism into a measure requires judging, task by task, what the technology can do. Successive generations of measures have gone about this differently: engineering bottlenecks read at the occupation-level, treating occupations as indivisible (Frey and Osborne 2017); linking AI applications to distinct human occupational abilities (Felten et al. 2021); the textual overlap of patents with job descriptions (Webb 2019); and scoring the task itself for its suitability for machine learning (Brynjolfsson et al. 2018). Large language models have since changed the capability of the measurement. Eloundou et al. (2024) have an LLM rate each O*NET task against a rubric, and show the ratings track expert judgment closely while covering the whole taxonomy in a way which no human exercise can afford. The approach now supports institutional studies. ILO work scores every occupation in the international classification to build a global exposure index (Gmyrek et al. 2023), the Tony Blair Institute and the OBR score the whole of O*NET for the UK (Office for Budget Responsibility 2025; Sharps et al. 2024), and the IMF's complementarity-adjusted index separates displacement from augmentation (Pizzinelli et al. 2023; Cazzaniga et al. 2024).

Each of these measures read exposure off different inputs such that they rank the same occupation differently, and so any estimate built on one inherits that disagreement. Recent evidence has also reassessed this problem by moving from a cross-measure concern into a within-measure one. Yin et al. (2026) score an identical task set with four LLMs and find average exposure differs by more than a factor of three across them, with downstream labour-market magnitudes varying roughly 2.4-fold by model. The divergence is systematic since each model carries a persistent tendency to score high or low, such that averaging does not remove it. Still, the acknowledgment of exposure scoring as a measurement exercise and not realised impact, is understated in institutional literature, along with a failure to propagate measurement uncertainty into the estimates built on top of it.

The geography of exposure and Scotland's composition

Exposure is a property of tasks and tasks bundle into occupations, meaning AI's incidence is uneven across space, though the literature disagrees on how just how uneven at the spatial scales of policy interest. Sharps et al. (2024) find limited regional variation in UK exposure, attributing it to the diversified sectoral mix of most regions. Additionally, the OECD finds cross-country exposure ranging from 16–77% with complementarity rising in regional education (OECD 2024), with European evidence finding automation risk driven by industrial structure and agglomeration (Crowley et al. 2021). The disagreement is partly artefactual since coarse administrative geographies average over the functional labour markets where local conditions operate (Overman and Xu 2022). Therefore, the regional signal appears most clearly in occupational mix and least clearly in coarse exposure averages. Consistent with this, Henseke et al. (2025) attribute the recent rise in UK exposure to shifts in occupational composition rather than to within-occupation task change.

Scotland's occupational mix is distinctive in ways that bear directly on exposure. Its public sector accounts for around 22% of employment against roughly 18% UK-wide, a gap persistent over time, with services accounting for 77.1% of Scottish GDP against 80.5% for the UK, with more local-government delivery of services such as social care and correspondingly less weight in the professional, scientific and information services concentrated in London and the South East (Scottish Government 2024; House of Commons

Library 2024). Financial, insurance and professional services are among the most exposed parts of the UK economy, and caring, elementary and skilled-trade work among the least (Office for Budget Responsibility 2025). These exposed sectors divide on the substitution–complementarity margin, since public administration, health and caring work scores high on complementarity and low on substitutability, whereas professional and financial work is exposed on both. Whether Scotland's lighter concentration in the latter harbours its workforce or merely denies it the productivity gains of complemented work is therefore ambiguous. The fiscal edge is sharper given that public-sector pay in Scotland sits about 5% above the UK median and has diverged upward since 2019 (Cribb et al. 2025), coupled against the backdrop of a chronic investment and productivity gap (Williams et al. 2025).

Gaps and open questions

The synthesis leaves several gaps. Exposure measures are reported as point estimates whose dependence on the scoring model and prompt is acknowledged but rarely quantified, and where a measure is carried into downstream exercises that institutions perform, exposure enters as though it were exact, with no way of telling if the resulting figures are trustworthy or where uncertainty arises. Thus far exposure for the UK is estimated only at coarse geographies and has not been posed as a compositional question for a devolved fiscal authority, so the Scotland–rUK differential, and which part of it survives the measurement fragility above, is absent from the literature. Finally, since no instrument for AI exposure yet matches the commuting-zone design of Acemoglu and Restrepo (2020), credible sub-national causal estimates are unavailable, so what remains tractable for this exercise is the measurement of the exposure differential and an audit of what the forecast implications drawn from it actually rest on. These gaps motivate three questions:

1. Which exposure estimands are robust to cross-model divergence? Specifically, do regional differentials survive the model-specific bias that destabilises absolute levels?

2. Does occupational composition leave Scotland more or less exposed than the rest of the UK, and where within that composition does the gap arise?

3. What does the exposure differential imply for the Scottish income tax net position, and which of the uncertainties along that chain govern the implication?

The remainder of this dissertation takes up these three questions in turn for the Scotland–rUK case.

3 Data

Occupational task content comes from O*NET, regional employment weights and worker-level earnings from the Annual Population Survey (APS), and a published crosswalk links the task data to the survey. I describe each in turn, deferring the construction of the exposure score itself, that is the scoring of tasks, the importance weighting, and the classification rule, to Section 4.

3.1 O*NET task content and the exposure score

O*NET is the occupational database maintained by the US Department of Labor, and is the most comprehensive task taxonomy available (the UK has no domestic equivalent).[1] For each O*NET–SOC occupation j the database records a set of task statements describing the discrete activities the work comprises, together with an Importance rating for each task on a five-point scale, measuring the degree to which a specific activity is required for a particular occupation. The task statements are the input the language model scores, and the importance ratings supply the weights $\omega_{j,t}$ in the occupation-level aggregate of Section 4. The cleaned file contains 17,945 occupation–task pairs across 894 occupations.

Now, O*NET records the tasks of an occupation, not of an occupation in a place, so a single score attaches to each occupation and carries no regional variation by construction. Additionally, because the scores attach to US occupations, they are of use to a UK study only once mapped onto the domestic classification, which is the role of the crosswalk.

3.2 Crosswalk to UK SOC 2020

The exposure scores are indexed by O*NET–SOC occupation, whereas the employment weights are recorded under UK SOC 2020. I bridge the two with a published concordance that maps O*NET–SOC 2019 codes to UK SOC 2020 unit groups, from which I form a continuous score E_k^{UK} for minor group k.[2] I work at the three-digit minor-group level, which trades a degree of occupational resolution for enough observations to support reliable regional employment shares once survey waves are pooled. Working at three digits averages away any Scotland–rUK sorting that happens within a minor group, just how a coarse geography compresses spatial variation. The public APS can't go finer than this, so the reported gap is to be seen as a lower bound. The concordance is many-to-many, so an aggregation rule is required; Appendix B states the rule I use and reports the differential under two alternatives.

3.3 Annual Population Survey employment weights

The APS is the Office for National Statistics' largest household survey, formed by pooling the quarterly Labour Force Survey with added regional samples, and is the standard source for sub-national labour-market statistics. It distinguishes individuals by country

[1] I use release 30.3, available at: `https://www.onetonline.org`

[2] Retrieved from: `https://www.nfer.ac.uk/media/1tojrw0o/matching_uk_soc2020_to_o_net.pdf`

and region, allowing Scotland to be observed directly and the rest of the UK (rUK) to be constructed as the aggregate of all other UK regions. From the APS I obtain weighted employment counts by SOC 2020 occupation for each region.

I pool four waves, 2022–2025, because a single wave leaves too many Scottish minor groups too thinly sampled for the employment shares to be precise, and the window is short enough that occupational structure is stable across it. The pooled employment counts $l_{k,r}^{\text{pool}}$ construct the regional employment shares $\eta_{k,r}$, and since exposure scores are common to all regions these shares are the only source of regional variation in the study.

The pooled shares remain survey estimates, with Scotland's thinner observation blocks carrying the larger sampling error. That error is independent of the scoring noise of Section 4.3 and, as Table 4 shows, several times larger in the differential, so I propagate both jointly and Table 4 reports each separately and in combination. Two features of the APS design, recorded in Appendix B, mean the resulting interval is not an exact coverage statement: the overlapping-wave structure that a person-level bootstrap treats as independent draws, and the 2022–24 fall in LFS response rates.

3.4 Earnings

Worker-level earnings also come from the pooled APS waves. The dependent variable is the APS derived gross hourly pay for employees, recorded alongside the occupation, industry, region, age, sex, full-time status, highest qualification and public-sector indicator that the wage regression of Section 4.4 uses. The earnings and employment results rest on a single consistent source, and, since the APS is public, the wage regressions reproduce without a restricted-data agreement.[3]

4 Methodology

I build an exposure score for every occupation from language-model ratings of its constituent tasks, then weight those scores by Scottish and rUK employment. The scores are the novel input, so most of this section concerns how they are constructed and how far they can be trusted; the labour-market accounting that follows is comparatively standard.

4.1 Exposure scoring

The OBR score each O*NET task from 0–100 for its amenability to AI over a ten-year horizon, separating tasks AI could substitute for from those it would merely complement. They classify an occupation as exposed where enough of its task content clears some

[3] The employer-reported Annual Survey of Hours and Earnings (ASHE), securely accessed from the UK Data Service, is the more accurate earnings source, since pay is returned by employers rather than recalled by respondents.

threshold. I adopt that architecture, its scoring scale and its substitute–complement distinction. However, their object is a single national figure with mine the difference between two regions. Additionally, they report point estimates without quantifying their dependence on the scoring framework, whereas I treat the scores as model-specific, only resting the regional claim on the component of the measurement that survives a change of model.

Procedure. I score the task content of every O*NET–SOC occupation j. For each task $t \in T_j$ the model returns two integer scores on $[0, 100]$: a substitution score $s^{\text{sub}}_{j,t}$, capturing the extent to which AI could perform the task without human involvement over the coming decade, and a complementarity score $s^{\text{comp}}_{j,t}$, capturing the extent to which it could raise a worker's productivity in the task without replacing them. The prompt sets the horizon, copies in task description from O*NET, and directs the model to potential considerations bearing on each judgment, such as whether the task requires physical presence, tacit empathy, or non-delegable accountability.

The two scores must then be combined into a single measure of how exposed a task is. For the exposure measurement I take the larger of the two, $s_{j,t} = \max(s^{\text{sub}}_{j,t}, s^{\text{comp}}_{j,t})$. I choose the maximum for commensurability, since it is the rule behind the OBR's single-amenability banding. That said, it is admittedly coarse, treating a task scored $(90, 10)$ as no less exposed than one scored $(90, 90)$, and as the maximum of two noisy scores it carries a small upward bias growing with the framing noise (Clark 1961). This is mitigated, firstly because the same rule applies to each region, so by Proposition 1 its effect largely cancels in the differential. Which I verify by reporting the gap under the substitution score alone and under an alternative $s_{j,t} = s^{\text{sub}}_{j,t} + \left(1 - s^{\text{sub}}_{j,t}/100\right) s^{\text{comp}}_{j,t}$ which treats complementarity as exposing only the share of a task left after substitution. Secondly, since substitution and complementarity have opposite-signed effects, they enter separately in Section 4.4.

Task scores then aggregate to the occupation. The occupation-level score is the importance-weighted mean of its task scores,

$$E_j \;=\; \frac{\sum_{t \in T_j} \omega_{j,t}\, s_{j,t}}{\sum_{t \in T_j} \omega_{j,t}} \;\in\; [0, 100], \tag{1}$$

with O*NET importance weights $\omega_{j,t}$, so that an occupation's exposure is reflective of its most central tasks.

Classification converts the continuous scores into the exposed and unexposed categories. A task is substituted where $s^{\text{sub}}_{j,t} \geq \theta^{\text{sub}}$, complemented where $s^{\text{comp}}_{j,t} \geq \theta^{\text{comp}}$ and the task is not substituted, and unexposed otherwise. My baseline thresholds are adopted from Office for Budget Responsibility (2025), $(\theta^{\text{sub}}, \theta^{\text{comp}}) = (70, 40)$, applied at the task level.[4] Weighting these task classes by importance in the same way as Equation (1) gives

[4]Sensitivity of the classified indices to the threshold pair is reported at $(60, 30)$ and $(80, 50)$ alongside

Sub_j and Comp_j, the shares of an occupation's task content falling in each exposed class.

Then, because the scores attach to US occupations, I map E_j, Sub_j and Comp_j through the crosswalk onto UK SOC 2020 minor group k, yielding the score E_k and the corresponding shares Sub_k and Comp_k, and classify at that level. Minor group k is materially exposed where $\text{Sub}_k + \text{Comp}_k \geq \frac{1}{2}$, and an exposed minor group is substituted where $\text{Sub}_k > \text{Comp}_k$ and complemented otherwise. O*NET records the tasks of an occupation, not of an occupation in a place, so the same E_k applies in Scotland and the rUK.

4.2 Regional indices and the compositional structure of the gap

The empirical question at hand is whether Scotland's workforce sits more or less in the path of this technology than the rest of the UK's. I develop an employment-weighted exposure index for each region: the exposure of a region's average job. For region $r \in \{\text{Scot}, \text{rUK}\}$, let $\eta_{k,r} = l^{\text{pool}}_{k,r} \big/ \sum_{k'} l^{\text{pool}}_{k',r}$ be the pooled employment share of minor group k. I form a classified exposed-employment share and a continuous index,

$$\widehat{E}_r = \sum_k \eta_{k,r}\, \mathbf{1}[\text{Class}(k) \in \{\text{Sub}, \text{Comp}\}]\,, \qquad \bar{E}_r = \sum_k \eta_{k,r} \frac{E_k}{100} \;\in\; [0,1]. \tag{2}$$

The first answers what fraction of employment sits in materially exposed occupations. The second uses the full scores and is the object for which the formal results below are exact. The gap $\Delta\widehat{E} = \widehat{E}_{\text{Scot}} - \widehat{E}_{\text{rUK}}$, with its continuous form $\Delta\bar{E}$, is the object of interest.

A shift-share decomposition makes clear that any regional difference in an employment-weighted index has, in principle, two sources. The regions may distribute their employment differently across occupations (composition), or the same occupation may be differently exposed in the two places (within-occupation):

$$\Delta\widehat{E} = \underbrace{\sum_k \Big(\eta_{k,\text{Scot}} - \eta_{k,\text{rUK}}\Big)\, \mathbf{1}[\text{Exp}(k)]_{\text{rUK}}}_{\text{composition}} + \underbrace{\sum_k \eta_{k,\text{Scot}} \Big(\mathbf{1}[\text{Exp}(k)]_{\text{Scot}} - \mathbf{1}[\text{Exp}(k)]_{\text{rUK}}\Big)}_{\text{within-occupation}} \tag{3}$$

Since the same score applies on both sides of the border, the within-occupation term vanishes because they are identical, and thus the entire gap is therefore compositional. Implying that any Scotland–rUK differential must reflect Scotland's distinctive occupational mix.

A single assumption must be made for this composition reading. It is that the within-occupation task profile of each UK minor group is the same in both regions. The assumption is not that occupations are internally homogeneous; Henseke et al. (2025) show they are not, with the highest score dispersion in occupations where a generic task description admits both an exposed and an unexposed reading. It is the weaker claim that whatever

the baseline.

heterogeneity exists is distributed similarly in the two regions.

Assumption 1 (Region invariance)**.** Writing T for a worker's task bundle, k for their minor group and $R \in \{\text{Scot}, \text{rUK}\}$ for region, the conditional distribution of tasks is region-invariant,

$$\Pr(T \in A \mid k, R = \text{Scot}) \;=\; \Pr(T \in A \mid k, R = \text{rUK}) \quad \text{for all } A. \tag{4}$$

Internal homogeneity would require T to be degenerate given k. What I need is only Equation (4), so that whatever within-occupation heterogeneity exists is distributed identically on either side of the border. This is reasonable for within one national economy since the qualifications, technologies, and regulatory standards that define an occupation are set at the UK level. The assumption fails only if task content differs across regions in a way correlated with exposure. I cannot test this directly without region-specific task data, but the international evidence suggests such within-occupation differences are small relative to composition (Handel 2012; Autor and Handel 2013).

4.3 Scoring error and the robustness of the differential

An occupation's task exposure score can be wrong for a few reasons, and the design responds to each of them separately. The framing noise from making a meaning-preserving change to the prompt is what the calibration of Equation (9) measures. Additionally, a particular LLM carries a persistent tendency to over- or under-score, and that systematic bias (Yin et al. 2026) is what Proposition 1 addresses and the cross-model test of Section 5.5 probes. Aggregation and crosswalk error is addressed by the operator and threshold robustness, with sampling error in the employment weights handled the APS bootstrap.

The cross-model divergence documented by Yin et al. (2026) would be fatal to a study reporting absolute levels as objective measurements, but is, fortunately, far less threatening to a study of regional differences. Consider a score generated by model-M, decomposed as

$$s_j \;=\; \theta^\star_j + b^M_j + \varepsilon_j, \qquad \varepsilon_j \sim N\Big(0, \sigma^2_{g(j)}\Big), \tag{5}$$

where $\theta^\star_j$ is the unobserved cross-model consensus, b^M_j is model M's systematic bias (its persistent tendency to over or under-score occupation j), and ε_j is mean-zero framing noise indexed by $g(j)$ as defined in the calibration below. The two error terms require different treatment. The noise term ε_j can be reduced through averaging and simulation, whilst the model-specific bias b^M_j identified by Yin et al. (2026), cannot, affecting every absolute estimate a model produces. In the following I show how this component conveniently cancels in the regional differential.

Conditioning on the frozen model absorbs b^M_j into a model-specific target $\theta^M_j = \theta^\star_j + b^M_j$, leaving the classical measurement model $s_j = \theta_j^M + \varepsilon_j$, with group variances $\sigma^2_{g(j)}$ to be calibrated below. Writing the continuous UK score as $E_k^M = E_k^\star + b_k^M$, the cancellation result follows.

Proposition 1 (Common-mode cancellation)**.** *The model-induced error in the differential* $\Delta\bar{E}^M = \bar{E}^M_{\text{Scot}} - \bar{E}^M_{\text{rUK}}$ *is*

$$\Delta\bar{E}^M - \Delta\bar{E}^\star \;=\; \frac{1}{100}\sum_k \Big(\eta_{k,\text{Scot}} - \eta_{k,\text{rUK}}\Big)\, b_k^M. \tag{6}$$

Hence (i) a pure level shift $b_k^M = b^M$ *leaves the differential exact,* $\Delta\bar{E}^M = \Delta\bar{E}^\star$*; and (ii) mean bias is removed by construction, with only the part of the bias co-varying with the regional employment-share gap surviving.*

See Appendix A for the proof of Proposition 1.

The differential therefore retains only the part of the bias co-varying with the cross-region employment-share gap, which Section 5.5 measures. The same compositional weighting that zeroes the within-occupation term of Equation (3) removes the common-mode bias, so *the differential is robust where the levels are fragile.*

This proposition implies that for any set of scores generated by a particular model, the cancellation result applies, and it can be shown the bias is bounded by the Cauchy-Schwarz inequality:

Corollary 1 (Cross-model stability)**.** *For any two models M and M',*

$$\Delta\bar{E}^M - \Delta\bar{E}^{M'} \;=\; \frac{1}{100}\sum_k \Delta\eta_k\Big(b_k^M - b_k^{M'}\Big), \qquad \Big|\Delta\bar{E}^M - \Delta\bar{E}^{M'}\Big| \;\le\; \frac{1}{100}\Big\|\Delta\eta\Big\|_2\,\Big\|b^M - b^{M'}\Big\|_2, \tag{7}$$

where $\Delta\eta_k = \eta_{k,\text{Scot}} - \eta_{k,\text{rUK}}$ *and the bias differences are demeaned. The cross-model movement in the differential is zero under a common level shift and bounded by how aligned the bias difference is with the share gap. See Appendix A for full discussion.*

Now, Proposition 1 protects the differential only against an additive bias such as that in Equation (5). However, a model may instead disagree about the scale of the index, compressing or stretching the score distribution while preserving the ordering. The next corollary covers that case, which Section 5.5 shows is the one the data present.

Corollary 2 (Affine bias and scale-free differentials)**.** *Suppose model M's scores are affine in the latent up to an idiosyncratic term,* $E_k^M = a^M + c^M E_k^\star + b_k^M$ *with* $c^M > 0$*. Then*

$$\Delta\bar{E}^M \;=\; c^M\,\Delta\bar{E}^\star \;+\; \frac{1}{100}\sum_k \Delta\eta_k\, b_k^M. \tag{8}$$

Hence (i) by Proposition 1 the intercept a^M *cancels exactly, with the slope rescaling the differential proportionally rather than cancelling; (ii) writing* SD^M *for the employment-*

weighted standard deviation of $\{E_k^M\}$, the standardised differential $\Delta\bar{E}^M/\mathrm{SD}^M$ removes both a^M and c^M, exactly when $b_k^M \equiv 0$ and up to the residual term otherwise; and (iii) replacing each score by its employment-weighted percentile rank yields a differential invariant to any *strictly increasing transformation of the scale, of which the affine map is a special case. See Appendix A for full discussion.*

Moving forward it will be useful to define that a statement about the differential is *scale-free* when it does not depend on how widely the index happens to be spread under a given model. Corollary 2 offers two: dividing by the model's own dispersion keeps the differential comparable with other standardised gaps but only under affine transformations, whereas percentile ranks undo any monotone transformation at the price of discarding everything but order

Proposition 1 handles a model that reads the whole economy as more or less exposed, and Corollary 2 one that uses more or less of the available scale. That is, re-drawing the scores from a different model should move the levels, rescale the differential, and leave its sign and scale-free magnitude unchanged. I test this by re-scoring the full task set on two alternative models and comparing the cross-model stability of $\Delta\bar{E}$ against that of $\widehat{E}_{\mathrm{Scot}}$ and $\widehat{E}_{\mathrm{rUK}}$ individually.[5] If the levels move while the gap holds, the regional finding survives the critique that undermines the absolute numbers.

Calibration and propagation. Differencing handles the bias, but the noise still has to be measured. Framing noise is the variation in a score under prompt changes that leave its meaning intact, and I estimate its variance rather than assume a form for it. I draw $N = 90$ occupations, eighteen from each of five broad occupational families $g(j)$ obtained by collapsing major groups onto their leading digit, and re-score each one's full task set under the conditions of Table 1.[6] Four of the conditions are meaning-preserving perturbations of the frozen prompt and the disagreement among them identifies the noise. Writing K for that set, with $|K| = 4$,

$$\widehat{\sigma}_j^2 = \frac{1}{|K|-1}\sum_{k\in K}\left(s_j^{(k)} - \bar{s}_j\right)^2, \qquad \widehat{\sigma}_g^2 = \frac{1}{|J_g|}\sum_{j\in J_g}\widehat{\sigma}_j^2, \tag{9}$$

where J_g collects the randomly sampled occupations of family g. The remaining two conditions, a five-year horizon and a single composite score, change the question itself and are reported as sensitivities.

I propagate this variance into the continuous aggregates by simulation. Across $R =$ 1,000 draws I perturb each occupation score, $s_j^{(r)} \sim N\left(s_j, \widehat{\sigma}^2_{g(j)}\right)$ truncated to $[0, 100]$, re-aggregate to UK minor groups, re-weight, and record $\Delta\bar{E}^{(r)}$, reporting the 95% interval

[5] I use one model from a different provider (OpenAI's GPT-4o) and one from a different tier of the same provider (Anthropic's Claude Haiku 4.5).

[6] Ten per family are drawn at random to estimate the variance and eight nearest the classification boundary are held out for the stability check and Appendix D sets out the partition reasoning.

Table 1: Calibration design: scoring conditions and the role of each

Variant	What changes	Role
V0	Baseline prompt, 10-year horizon	noise (reference)
V1	Assessment criteria in reverse order	noise: order and anchoring
V2	Paraphrased rubric, identical content	noise: wording
V3	Task presented before occupation	noise: framing
V4	5-year horizon, else identical	estimand sensitivity: horizon
V5	Single composite score	estimand sensitivity: rubric form

The within-model variance is estimated from V0, V1, V2 and V3 on the random sample only.; V4 and V5 are reported as named sensitivities. Cross-model contrasts come from the full re-scores under the two alternative models.

and the share of draws keeping the sign. Because independent draws understate aggregate uncertainty if framing errors move together, I also allow errors clustered within occupational family, with intraclass correlation $\rho \in \{0, 0.2, 0.4\}$, the higher values being more cautious. Appendix A derives the inflation factor.

Alongside the scoring draws, I bootstrap the pooled employment shares over persons within region and year while combining the two independent sources of uncertainty to construct an interval for $\Delta\bar{E}$. To show their relative importance, I also report the scoring and sampling intervals separately.

The classified share is nonlinear at the thresholds since a disturbance to a task near a cut-off flips its class, and with it the occupation's. I therefore give each occupation a stability score, defined as the importance-weighted probability that its task content keeps its class under re-framing, constructed in Appendix D along with the landmark I flag against.

4.4 Empirical specifications

The productivity projection converts an exposure gap into the fiscally useful units, with the wage regression serving as the micro counterpart of the variance propagation, whilst providing the exposure–pay alignment the translation needs.

The wage specification estimates the exposure–wage relationship at worker level,

$$\ln w_{i,t} = \alpha + \beta_1 E_{j(i)} + \beta_2 \, \mathrm{Scot}_i + \beta_3 \Big(E_{j(i)} \times \mathrm{Scot}_i \Big) + \gamma X_{i,t} + \delta_s + \delta_t + \varepsilon_{i,t}, \qquad (10)$$

where $\ln w_{i,t}$ is the log gross hourly pay of worker i in survey year t, $E_{j(i)} \in [0, 1]$ is the continuous exposure score of the worker's minor group $j(i)$, common to both regions and constant within an occupation, and Scot_i indicates residence in Scotland. The coefficient β_1 is the rUK gradient of pay in exposure, β_2 the Scotland wage-level shift, and β_3, the object of interest, the Scotland-specific rotation of that gradient. The controls $X_{i,t}$ are age, its square, sex and full-time status; δ_s and δ_t are industry-section and survey-year fixed effects; and observations are weighted by the APS income weight. Two further controls enter the richer specifications. Highest qualification is included as exposed occupations are graduate-heavy, so any Scotland–rUK difference in the within-occupation qualification profile would otherwise load onto β_3; and a public-sector indicator since published findings have shown Scotland's public sector workforce and pay have grown faster than the rest of the UK (Cribb et al. 2025). $E_{j(i)}$ varies only across occupations, every worker in a minor group shares one value of the regressor and the within-group correlation of residuals inflates conventional standard errors by the Moulton factor (Moulton 1990). I therefore cluster by occupation, the level at which the regressor varies.

The productivity projection in Section 6 translates the exposure differential into productivity, using the task-based framework of Acemoglu (2025) as parameterised by the Office for Budget Responsibility (2025). In that framework, AI raises total factor productivity by lowering the cost of the tasks it can perform, so the gain to a region depends on how much of its employment sits in exposed work. The proportional TFP gain for region r is

$$\Delta \ln \mathrm{TFP}_r \;\approx\; \kappa \sum_k \eta_{k,r}\, \pi_k\, \bar{E}_k. \tag{11}$$

Where π_k is the task-cost share and κ the calibrated average cost saving on exposed tasks. I import κ and the automatable fraction $\pi_k = \pi = 0.23$ from the OBR/Acemoglu calibration rather than estimating them, so the Scottish and rUK paths differ only through their employment-weighted exposure.

Corollary 3 (Downstream linear functionals)**.** *Let $F_r^M = \sum_k \eta_{k,r}\, q_k E_k^M$ with coefficients $q_k \geq 0$ common to both regions, and write $\bar{Q}_r = \sum_k \eta_{k,r} q_k$. Then*

$$\Delta F^M - \Delta F^\star \;=\; \sum_k \Delta\eta_k\, q_k\, b_k^M, \tag{12}$$

which under a pure level shift $b_k^M = b^M$ reduces to $b^M\left(\bar{Q}_{\mathrm{Scot}} - \bar{Q}_{\mathrm{rUK}}\right)$. Constant coefficients $q_k = q$ are therefore sufficient for exact cancellation. See Appendix A.

Equation (11) is simply the exposure index again, each score multiplied by a coefficient before the employment weights apply. Proposition 1 works because employment shares gaps sum to zero. Now, weighting each gap by q_k undoes that unless the coefficients are flat across occupations. Where they are not, a level shift survives, scaled by the cross-region difference in average loading. In this case, the calibration sets $\pi_k = \pi$ for every occupation, so $q_k = \kappa\pi$ is flat so by Corollary 3 each channel's TFP differential is a fixed multiple of $\Delta\bar{E}$, inheriting the cancellation exactly.

Appendix F derives Equation (11) and sources the calibration figures in full.

Table 2: Regional exposure indices and the Scotland–rUK differential

	Scotland	rUK	Gap	Relative gap
Panel A: continuous indices (index points, 0–100)				
Composite (max operator)	65.82	66.69	−0.86	−1.3%
Substitution only	38.30	38.65	−0.35	−0.9%
Complementarity	62.70	63.66	−0.96	−1.5%
Saturating combination	74.74	75.51	−0.77	−1.0%
Panel B: classified employment shares (%)				
Exposed	90.91	90.78	+0.14	+0.1%
Substituted	14.90	14.21	+0.68	+4.8%
Complemented	76.02	76.56	−0.55	−0.7%

Notes: Panel A is the continuous exposure index on a 0–100 scale and its gaps are in index points; Panel B reports shares of regional employment and its gaps are in percentage points of employment, under the central thresholds (substitution ≥ 70, complementarity ≥ 40).

5 Results

I first establish the measure on which the comparison rests, with the decomposition into a Scotland–rUK differential following. I then examine its sensitivity to the measurement choices that, according to Yin et al. (2026), undermine the interpretation of absolute exposure levels. Finally, the differential is translated into the quantities the SFC forecasts.

5.1 Occupational exposure

The baseline run scores all 17,945 task–occupation pairs across 894 O*NET occupations. Aggregated to the 104 UK SOC 2020 minor groups on which the regional comparison rests, E_k ranges from 31.7 (building finishing trades) to 84.3 index points (business, research and administrative professionals), with a cross-group standard deviation of 13.6 points. Reassuringly, then, the continuous measure carries the between-occupation variation a compositional comparison needs. The employment-weighted means, 66.7 points in rUK and 65.8 in Scotland, are high by the standards of the survey-based literature, and since the level is the object Section 4 argues is fragile to the scoring model, I place no weight on them.

The classified measure preserves little of this variation. Under the central thresholds, 84 of the 104 minor groups are Complemented, 9 Substituted and 11 Unexposed, so 90.8% of rUK and 90.9% of Scottish employment sits above the materiality cut. The classifier is in effect right-censored where once nine-tenths of employment in both regions is coded as exposed, any regional difference in the intensity of exposure is lost and the gap is driven by composition within the small unexposed tail. I therefore treat the continuous index as the primary measure.

5.2 The Scotland–rUK differential

Table 2 contains the headline exposure result. Scotland's employment-weighted continuous exposure is 0.86 index points below rUK on the 0–100 scale (65.82 against 66.69), a relative shortfall of 1.3%. The sign is stable across exposure operator: the substitution-only index gives −0.35 index points, the complementarity index −0.96, and the saturating combination −0.77. Scotland is therefore less exposed than the rest of the UK, by only a modest margin but with the sign well determined.

The scale-free form of the result is that Scotland sits about two percentile points below rUK in the UK occupational exposure distribution. The differential is small because the two regional labour markets are similar. Their employment-share vectors are close to collinear, with the largest difference across the 104 minor groups (functional managers and directors) amounting to only 1.6 percentage points of employment. Over the observed score range of 32–84 index points, no assignment of exposure can produce a large aggregate gap from compositional differences so small.

5.3 Anatomy of the differential

Table 3 reports the largest per-occupation contributions $\Delta\eta_k(E_k - \bar{E}_{\mathrm{rUK}})$ on each side, with the anatomy visualised into coarser groups in Figure 2. An occupation's contribution is the product of two signs: whether Scotland is over- or under-represented in occupation k, and whether that occupation sits above or below the rUK average exposure $\bar{E}_{\mathrm{rUK}}$ of 66.7. A contribution is negative either when Scotland is underweight in above-average work or when it is overweight in below-average work, and vice versa for positive cases.

Scotland is underweight in highly exposed professional occupations (functional managers and directors alone contribute −0.17 points and IT professionals −0.11) along with being overweight in service occupations that sit below the average, notably caring personal services (−0.17) and other elementary services (−0.16).

The offsetting positive contributions arise from Scotland being underweight in some low-exposure work, chiefly childcare and related personal services (+0.09) and construction and building trades (+0.06), whilst being overweight in a few high-exposure occupations, among them engineering professionals (+0.05), government administration (+0.05) and customer service (+0.04). The differential emerges from many small and economically coherent compositional differences rather than from a few influential occupations.

Nor is the differential solely a London effect. London's index is 69.30, compared with 66.33 for the UK excluding London, and thus Scotland's 65.82 remains 0.51 index points below the ex-London comparator. That is, excluding London closes 41% of the headline gap, but it does not eliminate it. The professional-management underweight persists (functional managers still contribute −0.14 points) as does the relative concentration in the care-sector. Scotland ranks fifth among the twelve UK nations and regions, behind

Table 3: Decomposition of the continuous differential: largest contributions

SOC	Minor group	E_k	$\Delta\eta_k$ (pp)	Contribution (pp)
113	Functional managers and directors	77.5	−1.58	−0.171
613	Caring personal services	51.2	+1.07	−0.165
926	Other elementary services	46.3	+0.76	−0.155
922	Elementary cleaning	39.1	+0.51	−0.140
213	IT professionals	80.9	−0.79	−0.112
611	Childcare and related personal services	47.5	−0.49	+0.094
531	Construction and building trades	37.4	−0.20	+0.059
212	Engineering professionals	79.3	+0.42	+0.053
411	Administrative: government and related	79.4	+0.39	+0.050
721	Customer service occupations	79.4	+0.35	+0.044

Notes: Contributions are $\Delta\eta_k(E_k - \bar{E}_{\mathrm{rUK}})$, in index points, summing exactly to the composite gap of −0.86 index points; the five largest on each side are shown. E_k is on the 0–100 exposure scale and $\Delta\eta_k$ is the Scotland-minus-rUK difference in the within-region employment share, in percentage points of employment.

London, the South East, the East of England, and the North West. The appropriate conclusion is not that Scotland is unusually insulated from AI exposure, but rather that London is unusually exposed, with Scotland sitting slightly below the middle of a distribution that spans barely 2.5 index points once London is excluded.

Expressed as a regression within one-digit occupational major groups, the same pattern holds on both the log-employment and employment-share margins (Appendix F), so the gap is not solely a matter of Scotland carrying fewer of the broad occupational families in which exposed work concentrates.

5.4 Scoring noise, sensitivity, and the composed interval

The calibration experiment delivers the within-model noise parameters. Across the meaning-preserving variants (V1–V3 of Table 1), the standard deviation of an occupation's total score ranges from 1.3 to 2.4 index points across the five occupational families, with task-level standard deviations of 3.4–5.1 index points for substitution and 3.6–6.0 for complementarity. Now, the estimand variants shift levels by much more. Shortening the horizon to five years (V4) moves composites by −1.5 index points on average, and collapsing the two-dimensional rubric to a single score (V5) moves them by −11.6. Interestingly, the cross-occupation correlations with the V0 baseline remain 0.998 and 0.956, so changing the question moves every level but barely moves the ordering of occupations. That is the same common-mode structure Proposition 1 formalises, with levels highly specification-dependent and relative positions not.

Table 4 propagates the noise, with each draw perturbing the occupation scores once and feeding both regions the same perturbed vector, such that scoring noise is common-

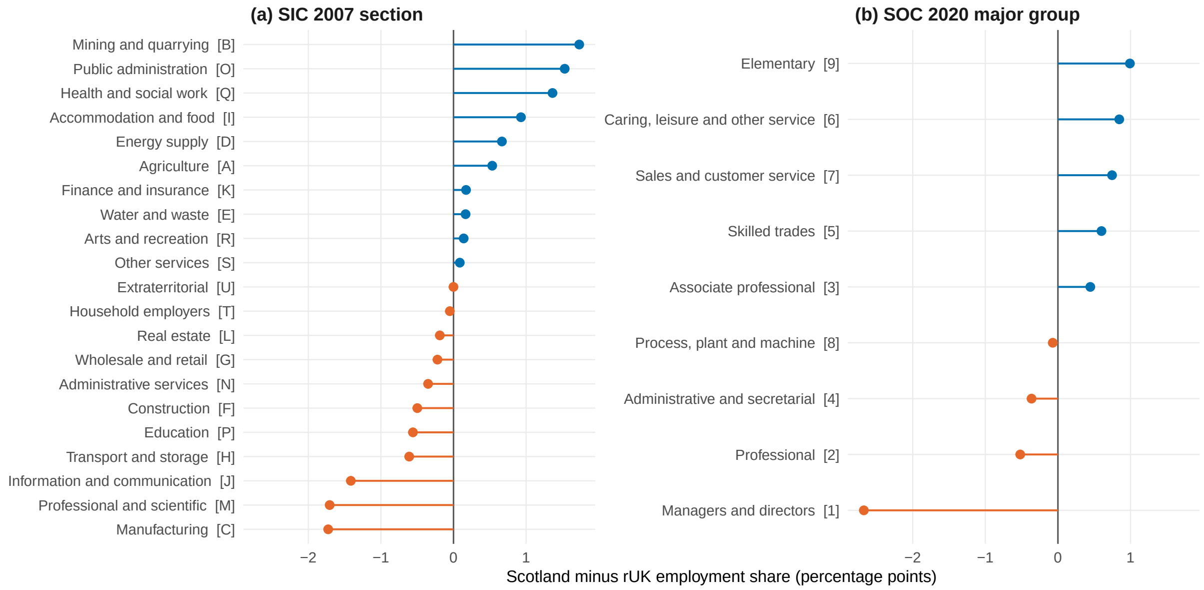

Figure 2: Orange indicates relative employment shortfall, with blue relative excess.

Table 4: Propagated uncertainty in the differential $\Delta\bar{E}$

Component	ρ	Mean	SD	2.5%	97.5%
Scoring only	0	−0.863	0.017	−0.897	−0.830
Scoring only	0.2	−0.864	0.031	−0.922	−0.801
Scoring only	0.4	−0.864	0.039	−0.938	−0.788
Sampling only	–	−0.864	0.099	−1.070	−0.690
Composed	0	−0.863	0.101	−1.069	−0.688
Composed	0.2	−0.862	0.103	−1.086	−0.677
Composed	0.4	−0.865	0.108	−1.091	−0.675

Notes: Index points on the 0–100 exposure scale; 1,000 draws per row. Scoring draws perturb occupation composites with the calibrated σ_g, correlated within occupational family at ρ; sampling draws bootstrap the employment shares over persons within region and year.

mode across regions by construction, and its near-total cancellation in the differential. Thus, the scoring rows are a numerical illustration of Proposition 1 and do not quantify uncertainty in the gap. The relevant uncertainty comes from sampling. Bootstrapping the employment shares widens the SD to 0.099 index points, and the composed interval at the most conservative within-family correlation is $[-1.09, -0.68]$ index points, with all 1,000 composed draws negative. Together this frames the headline exposure result as $\Delta\bar{E} = -0.86$ index points, with 95% interval $[-1.07, -0.69]$, most of whose width comes from survey sampling.

5.5 Cross-model stability of the differential

As Yin et al. (2026) show, choice of language model introduces a source of error that averaging cannot remove and therefore provides a direct test of Proposition 1. I re-scored

Table 5: Cross-model stability of the differential

Scoring model	Contrast	Mean bias	$\Delta\bar{E}$	$\text{corr}(b_k^M, \Delta\eta_k)$
Claude Sonnet 4.6	baseline	—	-0.86	—
Claude Haiku 4.5	same provider, lower tier	3.5	-0.53	-0.24
GPT-4o	different provider	9.1	-0.43	-0.23

Notes: Index points of the 0–100 scale. "Mean bias" is the mean absolute occupation-level score difference from the Sonnet baseline; $\Delta\bar{E}$ is the Scotland–rUK differential recomputed from each model's scores; b_k^M is the minor-group score difference from baseline, in index points, and $\Delta\eta_k$ the Scotland–rUK employment-share gap, and so the final column is the empirical surviving term in Equation (6).

all task–occupation pairs on two additional models, GPT-4o (a different provider) and Claude Haiku 4.5 (a lower tier of the same provider), and recomputed the differential from each.

The levels behave as predicted, with the alternative models disagreeing with the Sonnet baseline about absolute exposure by a mean occupation-level bias of 3.5 index points for Haiku and 9.1 for GPT-4o, with the latter divergence being of the order Yin et al. documents across frontier models. As anticipated, the differential, however, moves far less. Against a baseline gap of -0.86 index points, Haiku returns -0.53 and GPT-4o -0.43, so the differencing removes much, but not all, of the model disagreement. That much is Proposition 1 doing its work on the additive component of the bias.

By Corollary 2, we can say the cross-model differences are systematic rather than idiosyncratic. In both alternative models the bias is strongly correlated with the exposure level itself (0.88 for Haiku, 0.95 for GPT-4o), implying a compression of the scale rather than a simple shift.[7] The employment-weighted standard deviation of the minor-group scores falls from 13.58 index points under Sonnet to 9.59 under Haiku and 7.55 under GPT-4o, implying affine slopes of $c^M = 0.71$ and 0.56. Equation (8) then predicts differentials of -0.61 and -0.48 from compression alone, against realised values of -0.53 and -0.43, so the affine term accounts for roughly three-quarters of the movement in the Haiku gap and nine-tenths in the GPT-4o gap. Compression pulls the highest-exposure occupations down and Scotland is underweight in exactly those, ultimately reading as a 'smaller' differential.

Parts (ii) and (iii) of Corollary 2 tell us that dividing by each model's score dispersion removes any affine rescaling and percentile ranks remove any monotone one. Under these transformations the raw gaps of -0.86, -0.53, and -0.43 index points become -0.064, -0.055, and -0.057 standard-deviation units and -1.97, -1.76 and -2.01 percentile points (Table 6). The two-fold disagreement in index points collapses to about 13%, and GPT-4o, whose raw gap lies outside the within-model interval of Table 4, lands within

[7] This is a non-classical, mean-reverting error of the kind validation studies of survey data routinely find (Bound and Krueger 1991), with the difference that here there is no absolute truth to revert towards, only a baseline that is itself a model.

Table 6: Scale-free cross-model differentials

Model	Score SD (index pts)	Raw gap (index pts)	SD-unit gap (SDs)	Percentile gap (pctile pts)
Claude Sonnet 4.6 (baseline)	13.582	-0.863	-0.0636	-1.966
Claude Haiku 4.5	9.593	-0.531	-0.0554	-1.756
GPT-4o (2024-11-20)	7.549	-0.431	-0.0571	-2.008

Notes: The score SD is the UK-employment-weighted standard deviation of the 104 minor-group scores under each model. The SD-unit gap divides the index-point gap by that dispersion, removing any affine rescaling; the percentile gap replaces each score with its UK-employment-weighted percentile rank, removing any monotone rescaling.

2% of the baseline in percentile terms. These standardised gaps measure how much of the disagreement concerns Scotland's position within the distribution, and on that the three models are close.

The sign, the compositional structure, and the scale-free magnitude of about two percentile points all survive a change of scoring model, with Corollary 2 guaranteeing the last against any monotone rescaling. Conversely, the index-point does not survive, ranging from -0.43 to -0.86 because the models disagree about the dispersion of the index far more than about Scotland's position within it. I therefore report the scale-free magnitude as the headline, with -0.86 index points as the baseline model's expression of it.

5.6 Wage gradients: propagation at the worker level

The wage regression one, serves as the micro counterpart of the aggregate propagation, and two, supplies the exposure–pay alignment so the estimated Scottish gradient is an input to the fiscal translation rather than a free-standing result.

Higher-exposure occupations pay markedly more, reflecting how exposure aligns with professional and cognitive task content. The coefficient is 1.276 (SE 0.170), so each ten index points of exposure carries about 13.6% higher pay, implying that pay differs by 95.6% across the observed range of 31.7 to 84.3 index points. The Scotland interaction of -0.300 implies that the Scottish gradient is 0.976 against 1.276 in the rUK, close to a quarter flatter, indicating ten index points of exposure carry about 10.3% higher pay in Scotland, 3.3% lower than rUK. Within-occupation wages therefore rise less steeply with exposure in Scotland than in the rUK.

Adding highest qualification and a public-sector indicator (Table 7, column 2) cuts the exposure gradient from 1.276 to 0.941 but barely moves the Scotland interaction, so the flatter Scottish gradient is not a proxy for its qualification mix or its larger public sector. Giving London its own level and its own exposure slope (column 4) removes about three-tenths of the controlled interaction, so roughly a third of it is the capital and the remainder belongs to Scotland. Against the ex-London comparator the Scottish return to

Table 7: Exposure–wage regression

	(1) Baseline (full)	(2) +Controls	(3) +Region FE	(4) +London slope
Exposure E	1.276***	0.941***	0.913***	0.884***
	(0.170)	(0.144)	(0.143)	(0.143)
Exposure × Scotland	−0.300***	−0.269***	−0.216***	−0.187***
	(0.065)	(0.062)	(0.056)	(0.058)
Scotland	0.204***	0.173***		0.138***
	(0.044)	(0.042)		(0.040)
Exposure × London				0.354***
				(0.059)
London				−0.118***
				(0.039)
Demographic controls[a]	Yes	Yes	Yes	Yes
Added controls[b]		Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Region (GOR) FE			Yes	
Clustering	Occ.	Occ.	Occ.	Occ.
Sample	Full	GOR	GOR	GOR
Observations	176,444	176,024	176,024	176,024
Within R^2	0.174	0.235	0.226	0.243

Dependent variable: log gross hourly pay.
[a] Age, age^2, sex, and full-time/part-time status.
[b] Highest qualification (education) and a public-sector indicator.
Standard errors are in parentheses: $^{*}\,p < 0.1$, $^{**}\,p < 0.05$, $^{***}\,p < 0.01$.

exposure is roughly a fifth lower than in the rest of the UK (0.697 against 0.884), paying roughly 8% more in the least exposed occupations, drawing level at 73.8 index points, and about 2% less at the most exposed end. Scotland has less exposed employment and pays less for the exposed work it has, both of which compound in the fiscal translation of Section 6.2.

The wage regression carries a variable with error because the exposure scores are estimates, and random error in a regressor pulls its coefficient towards zero. How much depends on the noise relative to the real variation in exposure across occupations, and here scores run from 31.7 to 84.3 while the uncertainty about any one of them is a small fraction of that ($\lambda = 0.016$). The correction is accordingly 1.6%: the gradient moves from 1.276 to 1.297 and the Scotland interaction from −0.300 to −0.305, and Appendix E shows even that to be an upper bound. Scoring noise is immaterial here, as Table 4 finds it to be in the aggregate, with the largest uncertainty component from survey sampling.

One caveat on the wage exercises arises since gross hourly pay is recorded on the APS for employees only. Therefore, the estimates describe the employee earnings distribution but exclude the self-employed, who are disproportionately concentrated in the professional occupations sitting at the top of the exposure distribution.

6 From Exposure to the Net Position

To assess their fiscal implications, the exposure scores must first be mapped into productivity and then into income tax revenue. The following exercise is merely an accounting translation under stated assumptions and not a forecast. The parameters are imported, from the OBR/Acemoglu calibration in the first stage and from the block grant adjustment rules in the second, so that the Scottish and rUK paths differ only through their employment-weighted exposure. Proposition 1 survives to the net position, and the uncertainty in the fiscal figure can be appraised by rescaling the intervals already estimated in Section 5.4.

6.1 Implications for projected productivity

The projection converts the exposure gap into the units a fiscal claim has to be made in. Since the baseline projection is a linear functional of the scores with region-common coefficients, by Corollary 3 its regional differential inherits the Proposition 1 cancellation, so the differential is proportional to $\Delta \bar{E}$ by construction.

Under the central adoption scenario the composite index implies a cumulative ten-year TFP contribution of 2.362% for rUK against 2.331% for Scotland, a differential of -0.031 percentage points (Table 8). Moving the average cost saving on an exposed task, κ, across its low and high settings respectively changes the differential to -0.025 and -0.036 points. The complementarity projection gives -0.034 percentage points and the substitution projection -0.013, falling either side of the projection from the composite gap.

Now, equation (11) asks for the share of an occupation's task content that is exposed, and I use the continuous index $\bar{E}_k$ in its place. $\bar{E}_k$ is an importance-weighted mean amenability rating, whereas the framework wants an importance-weighted share of task content above a threshold. Panel C therefore repeats the central projection on the latter, $\mathrm{Sub}_k + \mathrm{Comp}_k$, for each of the three threshold pairs. The differential holds its sign throughout, ranging from -0.023 to -0.054 points, so regardless of functional used, Scotland's projected productivity gain from AI adoption falls short of rUK's.

6.2 Fiscal translation

The second stage carries the projection into the income tax net position.

Table 8: Projected ten-year TFP contribution and the Scotland–rUK differential

	κ	rUK	Scotland	Differential
Panel A. Composite exposure index, by cost-saving scenario				
Low	0.124	1.902	1.877	−0.025
Central	0.154	2.362	2.331	−0.031
High	0.182	2.792	2.755	−0.036
Panel B. Single-channel projections, central scenario				
Complementarity only	0.154	2.255	2.221	−0.034
Substitution only	0.154	1.369	1.356	−0.013
Panel C. Exposed task-content share $\mathrm{Sub}_k + \mathrm{Comp}_k$*, central scenario*				
Thresholds $(60, 30)$	0.154	3.240	3.218	−0.023
Thresholds $(70, 40)$	0.154	2.953	2.911	−0.042
Thresholds $(80, 50)$	0.154	2.725	2.671	−0.054

Notes: Cumulative contribution to total factor productivity over the ten-year horizon, in per cent; the differential is Scotland minus rUK in percentage points. Panel B rows are alternative projections run on the two channel indices. Panel C replaces $\bar{E}_k$ with the importance-weighted share of task content classified as exposed. The band carried into Table 9 is the Panel A range.

Translating the TFP differential into a budgetary effect requires a sequence of accounting links. Equation (11) delivers a path for total factor productivity, whereas earnings track output per hour. Under constant returns with a capital share of one third, a permanent one per cent increase in TFP therefore raises output per hour by $1/(1-\alpha) = 1.5$ per cent once the capital stock has adjusted (Office for Budget Responsibility 2025). As standard in the literature, I then assume full long-run pass-through from labour productivity to real earnings, which, spread evenly across the horizon, implies about -0.0046 percentage points of annual earnings growth for the central setting. The earnings differential is then translated into tax revenue. With a progressive income tax schedule, the elasticity of non-savings, non-dividend (NSND) revenue with respect to earnings exceeds one. A permanent increase in earnings raises revenue more than proportionately in line with the ratio of the income-weighted average marginal tax rate to the average tax rate on total income — frozen higher, advanced and top-rate thresholds keep that ratio above one (Scottish Government 2026). I estimate on the APS earnings distribution under the prevailing Scottish schedule, the elasticity to be 1.92, which sits towards the upper end of the 1.7–2.0 range reported by international studies of progressive income tax systems (Appendix F.2). The final link is the block grant adjustment, which is indexed on a per-capita basis to rUK revenue growth, such that the net position responds to the Scotland–rUK differential in per-capita revenue growth (Scottish Government and HM Treasury 2023). This removes the component of the shock common to the two regions and so the differential, which carries the Scottish schedule and the Scottish base. Thus, only 1.92 enters, applied to Scottish NSND revenue of £18.6 billion in 2024–25 (HM Revenue and Customs 2026).

Table 9: Translation of the exposure differential into the income tax net position

	Low	Central	High
TFP differential (pp, 10-year cumulative)	−0.025	−0.031	−0.036
Labour productivity differential (pp, 10-year cumulative)	−0.037	−0.046	−0.054
Implied earnings growth differential (pp p.a.)	−0.0037	−0.0046	−0.0054
Implied NSND revenue growth differential (pp p.a.)	−0.0071	−0.0088	−0.0104
Net position sensitivity (£m p.a.)	−1.3	−1.6	−1.9
Memo: terminal annual effect, year ten (£m)	−13	−16	−19
Memo: cumulative effect over the ten years (£m)	−73	−90	−107
Memo: forecast net position 2026–27 (£m)		969	

Notes: The top row is the composite ten-year Scotland–rUK differential of Table 8, Panel A. The revenue base is Scottish NSND revenue of approximately £18.6 billion in 2024–25 (HM Revenue and Customs 2026) and the memo net position is the SFC January 2026 forecast for 2026–27 (Scottish Fiscal Commission 2026). Population growth differential is omitted throughout.

Together, this sequence gives a net-position sensitivity of about −£1.6 million a year at the centre, and −£1.3 to −£1.9 million across the cost-saving band (Table 9). Scottish income tax outturn is reconciled against forecast three years after the event, and the reconciliation is applied to the budget in the year it lands (Scottish Government and HM Treasury 2023). A differential driven by occupational structure is persistent, so it enters each year's forecast, is carried into that year's reconciliation, and is applied *ex post* against a budget already set three years earlier. Ultimately the level effect accumulates even if the annual increment does not grow, which is why the ten-year cumulative figure in Table 9 is far larger than any single year's, and why it arrives with a three-year lag. The terminal annual effect, in a decade, is £16 million (about 1.7 per cent of next year's forecast position).

Despite the small magnitude, I carry the noise from the differential through to the fiscal effect (Table 10). Framing noise is the smallest source at 9 per cent either side of the centre, because both regions receive the same perturbed score vector and Proposition 1 removes almost all of it. Sampling and the adoption scenario are comparable at around two-fifths each, so variation from importing calibration numbers is similar to that arising from the data. Choice of scoring model is the most significant at 50 per cent, and due to being systematic it cannot be shrunk (Yin et al. 2026).

Across every source the sensitivity stays between £0.8 and £2.1 million a year, at most a fifth of one per cent of the SFC's forecast net position of £969 million for 2026–27 (Scottish Fiscal Commission 2026), and well inside forecast-error ranges running to the hundreds of millions. Clearly, AI exposure is not a material differentiator in next year's forecast, but the direction and persistence of the differential remain relevant in a system that reconciles forecasts three years in arrears. Additionally, in another application where magnitude is non-trivial, there may be sources of uncertainty, chiefly the choice of LLM, that can have a significant impact on the figures that institutions lean on.

Table 10: Error in the annual net-position sensitivity

Source of uncertainty	Gap (index points)	£m p.a.	Span (% of central)
Framing noise, $\rho = 0.4$	-0.94 to -0.79	1.50 to 1.78	17
Adoption scenario (κ band)	*held at* -0.86	1.31 to 1.93	38
APS employment-share sampling	-1.07 to -0.69	1.31 to 2.03	44
Composed measurement interval	-1.09 to -0.68	1.28 to 2.07	48
Choice of scoring model	-0.86 to -0.43	0.82 to 1.63	50

Notes: Each row carries a range on the exposure differential through the chain of Table 9 against a central figure of £1.6 million.

7 Conclusions

This dissertation asked what a fiscal authority can learn from LLM-based exposure measures, given that they are demonstrably model-dependent. The answer is that it can learn the difference but not the level. Where the question is posed as a gap across regions, and the gap is read scale-free, the measure holds; levels carry a systematic bias that averaging cannot remove and will not support the analyses built on top of them.

Scotland sits about two percentile points below the rest of the UK in the UK occupational exposure distribution, and 0.86 index points on the baseline model's 0–100 scale. The composed 95% interval on that figure, $[-1.07, -0.69]$, has most of the width coming from survey sampling, and describes precision within one model rather than agreement across models. London accounts for about two-fifths of the gap, and Scotland ranks fifth of the twelve UK nations and regions; against the ex-London comparator the shortfall narrows to 0.51 index points, so the finding is less that Scotland is insulated than that London is unusually exposed. Behind the gap is a composition underweight in professional-management work, which scores high, and overweight in the care sector, which scores low, both of which persist against the ex-London benchmark.

The robustness of the differential is established mathematically and then verified through a re-scoring exercise. Proposition 1 shows the differential removes the common component of any model-specific bias, with Corollary 1 bounding what survives. Corollary 2 covers the case the data in fact present, where the alternative models compress the scale rather than reordering occupations, and since Proposition 1 does not protect against this, a standardised or rank-based differential is supported. Specifically, I find on the re-score that the raw gap roughly halves while absolute levels move by up to 9.1 index points, three-quarters to nine-tenths of that movement being the affine rescaling of Corollary 2. The remainder is small in the terms Corollary 1 sets: the model bias correlates with the employment-share gap at -0.23 and -0.24, so the non-affine residual reaches only about a quarter of the Cauchy–Schwarz worst case. The sign of the differential, its composition, and its scale-free magnitude are all robust to the choice of scoring model.

Carried into productivity projections à la Acemoglu (2025), the gap implies a ten-year

TFP differential of -0.031 percentage points, or three basis points, which translates into a net-position sensitivity of about £1.6 million a year. Against the £969 million forecast net position, this figure is modest and well inside the SFC's own income tax forecast-error ranges, but it is persistent, and applies afresh at each three-year reconciliation.

Since it is a theme of this paper not to trust levels, it would be remiss to report a fiscal translation figure and abstract from model dependency. Every step from score index to the net position multiplies by a coefficient common to the two regions, thus by Corollary 3 making the chain a linear functional: the Proposition 1 cancellation reaches the net position, and its uncertainty can be priced by rescaling. Doing so orders the sources. Scoring noise moves the annual figure least, by 9 per cent either side of the centre; survey sampling and the adoption scenario shift it by around two-fifths of the central figure each; and the choice of scoring LLM by 50 per cent, which averaging cannot reduce. The wage gradient makes the same point at worker level, where the errors-in-variables correction is 1.6%, changing nothing.

For a forecasting body the guidance is short. Use continuous indices in place of classified shares, because a classifier that codes 91% of employment as exposed censors any variation required for regional comparison. Trust differentials over levels, as differencing removes the bias that makes levels a property of the scoring model rather than of the labour market. And expect the choice of model to move magnitudes. Design around it: report the sign and the percentile statement, neither of which inherits the model's scaling, and let direction and persistence take precedence over the index-point figure.

The limitations are as follows. Since no instrument for AI exposure yet matches the commuting-zone design of the automation literature, nothing here is causally identified, and so what I report are compositional measurements rather than displacement effects. Assumption 1, region invariance of within-occupation task content, cannot be tested without UK regional task data that no source provides, though the industry-margin diagnostics of Appendix B bound the strain it is under. Lastly, the fiscal translation is merely an accounting exercise using revenue elasticity computed from the Scottish schedule rather than assumed, with the pass-through from productivity to earnings imposed, and the population channel running through per-capita indexation is left out entirely.

What the framework offers extends beyond this application. Any exposure-based regional estimand inherits the same common-mode structure, for any pair of regions and any scoring model: an employment-weighted contrast whose weights sum to one differences out whatever the model does uniformly, and what remains is bounded by the alignment of the model's bias with the share gap. And where magnitude is not immaterial as it is here, the choice of scoring model can impact the figures institutions lean on even more than the data do.

The natural extensions centre around data this project could not obtain. Employer-reported ASHE earnings under a secure Data Access Agreement would replace recalled pay and supply an external check on the pooled occupational shares. The secure-access APS would support the four-digit differential the public file compresses, along with unit-group employment weights for the second step of the crosswalk.

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A Proofs and Derivations

A.1 Common-mode cancellation (Proposition 1)

Proof. Recall the model-M continuous region index

$$\bar{E}_r^M = \frac{1}{100} \sum_k \eta_{k,r} \, E_k^M, \qquad E_k^M = E_k^\star + b_k^M, \tag{13}$$

where $E_k^\star$ is the latent exposure of minor group k and b_k^M the model-M bias. Substituting the second expression into the first and separating the latent and bias components,

$$\bar{E}_r^M = \frac{1}{100} \sum_k \eta_{k,r} E_k^\star + \frac{1}{100} \sum_k \eta_{k,r} b_k^M = \bar{E}_r^\star + \frac{1}{100} \sum_k \eta_{k,r} b_k^M. \tag{14}$$

Differencing Equation (14) across regions $r \in \{\text{Scot}, \text{rUK}\}$, the latent parts collapse to $\Delta \bar{E}^\star = \bar{E}_{\text{Scot}}^\star - \bar{E}_{\text{rUK}}^\star$ and the bias parts to

$$\Delta \bar{E}^M - \Delta \bar{E}^\star = \frac{1}{100} \sum_k \Big(\eta_{k,\text{Scot}} - \eta_{k,\text{rUK}} \Big) b_k^M = \frac{1}{100} \sum_k \Delta \eta_k \, b_k^M, \tag{15}$$

which is Equation (6). The key step is that the employment shares are proper weights in each region, $\sum_k \eta_{k,r} = 1$, so that the share gaps sum to zero:

$$\sum_k \Delta \eta_k = \sum_k \eta_{k,\text{Scot}} - \sum_k \eta_{k,\text{rUK}} = 1 - 1 = 0. \tag{16}$$

Because of Equation (16), adding or subtracting any occupation-invariant constant $\bar{b}^M$ inside the sum in Equation (15) leaves it unchanged:

$$\frac{1}{100} \sum_k \Delta \eta_k \, b_k^M = \frac{1}{100} \sum_k \Delta \eta_k \Big(b_k^M - \bar{b}^M \Big) + \underbrace{\frac{\bar{b}^M}{100} \sum_k \Delta \eta_k}_{=0} = \frac{1}{100} \sum_k \Delta \eta_k \Big(b_k^M - \bar{b}^M \Big). \tag{17}$$

Taking $\bar{b}^M$ to be the mean model bias, Equation (17) shows the differential error to be the covariance between the cross-region share gap and the demeaned bias. Part (i) of Proposition 1 follows on setting $b_k^M = b^M$ for all k, whereupon $b_k^M - \bar{b}^M = 0$ and the right-hand side vanishes; the absolute levels of Equation (14) remain contaminated by $\frac{1}{100} \sum_k \eta_{k,r} b^M$, of the order of the mean bias. Part (ii) is Equation (17) itself. $\square$

A.2 Cross-model stability (Corollary 1)

Applying Equation (15) to two models M and M' and differencing the two error expressions, the latent benchmark $\Delta\bar{E}^{\star}$ cancels, leaving

$$\Delta\bar{E}^{M} - \Delta\bar{E}^{M'} = \frac{1}{100}\sum_{k}\Delta\eta_k\Big(b_k^{M} - b_k^{M'}\Big). \tag{18}$$

By the same zero-sum argument of Equation (16), Equation (18) is invariant to a common level shift in either model's bias and is therefore zero whenever the cross-model bias difference $b_k^M - b_k^{M'}$ is constant in k. More generally it is bounded, by the Cauchy–Schwarz inequality, by

$$\Big|\Delta\bar{E}^{M} - \Delta\bar{E}^{M'}\Big| \;\le\; \frac{1}{100}\,\Big\|\Delta\eta\Big\|_2\,\Big\|b^{M} - b^{M'}\Big\|_2, \tag{19}$$

so that the cross-model movement in the differential is small whenever the share gap and the bias difference are weakly aligned, even when the bias difference itself is large. This is the formal content of the claim that the differential is more stable across models than the levels, and together Equations (18) and (19) prove Corollary 1.

A.3 Affine bias and scale-free differentials (Corollary 2)

Suppose model M is allowed three ways of being wrong at once: it may shift the origin of the scale, stretch or compress it, and on top of that mis-score individual minor groups. Formally, let $E_k^M = a^M + c^M E_k^{\star} + b_k^M$ with $c^M > 0$. Substituting into the region index and differencing as in Equations (14) and (15),

$$\Delta\bar{E}^{M} = \frac{1}{100}\sum_{k}\Delta\eta_k\Big(a^M + c^M E_k^{\star} + b_k^M\Big) = \underbrace{\frac{a^M}{100}\sum_{k}\Delta\eta_k}_{=0} + c^M\,\Delta\bar{E}^{\star} + \frac{1}{100}\sum_{k}\Delta\eta_k\, b_k^M, \tag{20}$$

which is Equation (8). The intercept vanishes by Equation (16), exactly as the mean bias does in Proposition 1 since an affine model's additive part is a common-mode error. The slope, however, is not. It multiplies the differential, so a model that uses less of the $[0, 100]$ scale returns a proportionally smaller gap in index points even where it agrees perfectly about which occupations are the exposed ones. Thus, bias b_k^M co-varies with the exposure level and therefore, in part, with the share gap.

Part (ii) follows because the transformation is applied to the score distribution, which is common to the two regions. Writing SD^M for the employment-weighted standard deviation of $\{E_k^M\}$ across the UK, a compression by c^M shrinks the gap and the dispersion in the same proportion. So when $b_k^M \equiv 0$ we have $\mathrm{SD}^M = c^M\mathrm{SD}^{\star}$, and the two factors cancel:

$$\frac{\Delta\bar{E}^{M}}{\mathrm{SD}^{M}} = \frac{c^M\Delta\bar{E}^{\star}}{c^M\mathrm{SD}^{\star}} = \frac{\Delta\bar{E}^{\star}}{\mathrm{SD}^{\star}}, \tag{21}$$

and neither a^M nor c^M survives. With $b_k^M \neq 0$ the standardised differential retains $\left(\frac{1}{100}\sum_k \Delta\eta_k b_k^M\right)/\mathrm{SD}^M$, which is the non-affine residual and is what Table 6 measures.

Part (iii) is immediate. A percentile rank depends on the scores only through their ordering. Any strictly increasing transformation, of which $E \mapsto a^M + c^M E$ with $c^M > 0$ is one, cannot change an ordering. The percentile differential is therefore untouched by any monotone rescaling, not just an affine one. The price is that it throws away cardinal information of the index-point gap. That is, it can say Scotland is two percentile points down, but not that the gap is twice as large as some other gap.

A.4 Downstream linear functionals (Corollary 3)

Proof. Recall $F_r^M = \sum_k \eta_{k,r}\, q_k E_k^M$ with coefficients $q_k \geq 0$ common to both regions. Substituting $E_k^M = E_k^\star + b_k^M$ and differencing across regions gives Equation (12). Unlike the unweighted case of Equation (16), the weighted share gaps need not sum to zero: $\sum_k \Delta\eta_k q_k = \bar{Q}_{\mathrm{Scot}} - \bar{Q}_{\mathrm{rUK}}$ with $\bar{Q}_r = \sum_k \eta_{k,r} q_k$. Under a pure level shift $b_k^M = b^M$, Equation (12) therefore reduces to $b^M\left(\bar{Q}_{\mathrm{Scot}} - \bar{Q}_{\mathrm{rUK}}\right)$. When the coefficients are constant, $q_k = q$, the sum collapses to $q\, b^M \sum_k \Delta\eta_k = 0$ by Equation (16), and the cancellation is exact. □

The continuous exposure gap is the special case $q_k = 1/100$; the baseline TFP differential of Section 4.4 is the special case $q_k = \kappa\pi_k$ with $\pi_k = \pi$ constant across occupations.

A.5 Errors-in-variables attenuation

Consider the conditional wage regression Equation (10) in the single-regressor form $\ln w_i = \alpha + \beta_1 E_{j(i)} + \nu_i$, where the true continuous exposure $E_{j(i)} \in [0,1]$ is observed with classical within-model error $\tilde{E}_{j(i)} = E_{j(i)} + u_i$, $u_i \sim (0, \sigma_u^2)$ independent of $E_{j(i)}$ and ν_i. The within-model scoring variance is estimated on the $[0,100]$ scale as $\widehat{\sigma}^2_{g(j)}$, which rescaling to the $[0,1]$ index gives $\sigma_u^2 = \widehat{\sigma}^2_{g(j)}/10^4$. The OLS estimator of β_1 using $\tilde{E}$ has probability limit

$$\operatorname{plim} \widehat{\beta}_1^{\mathrm{OLS}} = \beta_1 \cdot \frac{\mathrm{Var}(E)}{\mathrm{Var}(E) + \sigma_u^2} = \beta_1(1-\lambda), \qquad \lambda \equiv \frac{\sigma_u^2}{\mathrm{Var}(E) + \sigma_u^2}, \tag{22}$$

the standard attenuation result for classical measurement error in a single regressor (Wooldridge 2010, ch. 4), where λ is the reliability complement and the survey-data context is that of Bound and Krueger (1991).

$$\widehat{\beta}_1^{\mathrm{EIV}} = \frac{\widehat{\beta}_1^{\mathrm{OLS}}}{1-\widehat{\lambda}}, \qquad \widehat{\lambda} = \frac{\widehat{\sigma}_u^2}{\widehat{\mathrm{Var}}(\tilde{E})}, \tag{23}$$

with standard errors obtained by the delta method on the smooth function $(\widehat{\beta}_1^{\mathrm{OLS}}, \widehat{\lambda}) \mapsto \widehat{\beta}_1^{\mathrm{OLS}}/(1-\widehat{\lambda})$. The correction to the Scotland interaction β_3 in the full specification

Equation (10) is treated as approximate, since the interaction regressor $E_{j(i)} \times \text{Scot}_i$ inherits the measurement error of $E_{j(i)}$ only on the Scottish subsample and the resulting attenuation is no longer a single scalar factor.

A.6 Correlated within-family errors in the Monte Carlo

The procedure of Section 4.3 draws $s_j^{(r)} \sim N(s_j, \widehat{\sigma}^2_{g(j)})$ across occupations j, independently under $\rho = 0$ and with an equicorrelated structure within each occupational family otherwise. Consider the contribution of a single family g containing n_g occupations to an employment-weighted aggregate, and let $W_g = \sum_{k \in g} \eta_{k,r}$ be the employment weight the family carries. The family's occupation-level contribution to the variance, under independent draws, is $W_g^2 \widehat{\sigma}_g^2 / n_g$. Imposing $\text{Cov}(\varepsilon_i, \varepsilon_{i'}) = \rho\, \widehat{\sigma}_g^2$ for $i \neq i'$ within the family gives

$$\text{Var}\Big(\bar{E}_r\Big) = \sum_g \frac{W_g^2\, \widehat{\sigma}_g^2}{n_g} \underbrace{\Big[1 + (n_g - 1)\rho\Big]}_{\text{design effect}}, \tag{24}$$

which reduces to the independent case at $\rho = 0$ and rises to $W_g^2 \widehat{\sigma}_g^2$ at $\rho = 1$, where the family moves as one. Notice, the bracket (the familiar design-effect factor) exceeds one for any $\rho > 0$, so positively correlated framing errors widen the interval relative to the independent baseline. I report the propagation across $\rho \in \{0, 0.2, 0.4\}$ and read the spread across those rows as the sensitivity of the reported uncertainty to the assumption.

B Data Construction and Coverage

B.1 Crosswalk from O*NET–SOC to UK SOC 2020

The exposure scores are indexed by O*NET–SOC occupation and the employment weights by UK SOC 2020, so the two are bridged by a published concordance that maps O*NET–SOC 2019 codes to UK SOC 2020 unit groups. I work at the three-digit minor-group level. This trades a degree of occupational resolution for sample sizes large enough to support reliable regional employment shares once survey waves are pooled, which is the binding constraint for Scotland, where finer groupings thin out quickly. The coarser working level also makes the region-invariance assumption of Assumption 1 more comfortable, since averaging over a minor group smooths the fine distinctions along which any residual regional difference in task content would appear.

Aggregation rule. The concordance is many-to-many, so an aggregation rule is required. Each UK SOC 2020 unit group receives the mean of the scores of the O*NET occupations mapping into it, weighted by the concordance weights, and then aggregated to the three-digit minor group by unweighted mean. A consequence of the rule is that the unweighted second step gives a thinly populated unit group the same relevance as a majorly populated one. I therefore recompute the differential under two alternative rules, equal weights at both steps, and mapping-mass weights at the second step, which Table A1 reports alongside the baseline. Weighting the second step by UK unit-group employment would be stronger still, but the public APS does not publish employment below the minor group.

Table A1: Sensitivity of the differential to the crosswalk aggregation rule

Aggregation rule	Scotland	rUK	Gap
Baseline (weighted, then unweighted)	65.82	66.69	-0.8633
Equal weights at both steps	65.82	66.69	-0.8633
Mapping-mass weights at step 2	65.95	66.85	-0.8988

Notes: Continuous composite index under the maximum operator, in index points. The baseline row reproduces the headline gap of Table 2 by construction.

B.2 APS pooling and sample-count diagnostics

Pooling four APS waves (2022-2025) is necessary because a single wave leaves too many Scottish minor groups with too small a sample for the employment shares to be precise, and the window is short enough that occupational structure is stable across it. 23 of the 104 pooled Scottish minor groups contain fewer than 100 respondents across the four waves, with the smallest containing 11.

Two design features the bootstrap does not capture. The APS is assembled from overlapping Labour Force Survey waves, so a respondent can appear in two adjacent annual datasets. A person-level bootstrap of the pooled file treats those appearances as independent draws and ignores the survey's clustered design. Both simplifications understate the sampling variance of the shares, so the composed interval of Table 4 is not an exact coverage statement. The pooled window also spans the 2022–24 fall in LFS response rates, whose consequences for sub-national estimates the ONS has itself flagged. Differential non-response between Scotland and the rUK would shift the shares rather than widen their interval, and no bootstrap can detect it.

The industry margin of region invariance. Exposure attaches to an occupation, not to an occupation within an industry. So Assumption 1 fails if exposure varies across the industries in which an occupation is employed *and* Scotland's within-occupation industry mix differs from the rUK's. The APS records industry alongside occupation, so the second half of that condition is measurable. For each minor group k I compute the total-variation distance between the Scottish and rUK industry distributions within the occupation,

$$\mathrm{TV}_k = \tfrac{1}{2}\sum_i \Big|\Pr(i \mid k, \mathrm{Scot}) - \Pr(i \mid k, \mathrm{rUK})\Big|,$$

which is the share of the occupation's employment that would have to move between industries to make the two distributions coincide.

Divergence is modest. Across the 99 adequately populated minor groups the median TV is 0.14 and the ninetieth percentile 0.26 (Table A2). The Scotland-employment-weighted mean $B = \sum_k \sigma_{k,\mathrm{Scot}} \mathrm{TV}_k = 0.114$ turns this into a bound: if within-occupation exposure spans at most δ index points across industries, the error in $\Delta\bar{E}$ cannot exceed δB. The design cannot observe δ, so the bound is only as informative as the value one is willing to assume. A spread of five index points admits an error of 0.57 points against a gap of -0.86.

It is ambiguous where the strain sits exactly. TV is mildly negatively correlated with the absolute gap contributions (-0.19), so divergence does not concentrate in the minor groups that generate the differential; but it is mildly positively correlated with exposure itself ($+0.25$). The bound depends on TV interacted with employment-share divergence rather than on exposure alone, so the second correlation does not imply bias.

Table A2: Within-occupation industry divergence and the implied bias bound

Statistic	Value
Minor groups with adequate cells	99
Share of Scottish employment covered	0.9966
Median TV	0.1404
90th percentile TV	0.2606
Bias bound multiplier B	0.1143
Implied bound on gap, $\delta = 5$ index points	0.5713
Correlation of TV with exposure	0.2474
Correlation of TV with |gap contribution|	-0.1913

Notes: Computed on minor groups with at least thirty respondents in each region across the pooled 2022–2025 waves; coverage is reported as the share of Scottish employment those minor groups carry. $TV \in [0,1]$, with 0 indicating identical within-occupation industry composition on both sides of the border. The final two rows test whether the strain on Assumption 1 concentrates where the gap is generated.

B.3 Full industry contributions to exposure

Table A3: Industry-mix and within-industry contributions by SIC section

Industry	Share gap (Scot - rUK, pp)	Within-industry exposure gap	Industry-mix contribution	Within-industry contribution
Extraterritorial bodies	-0.0004503	-4.656	-0.00002685	-0.009024
Household employers	-0.0504	-2.141	0.008661	-0.002386
Water & waste	0.1663	-1.939	-0.002096	-0.01772
Arts & recreation	0.1401	-1.372	-0.001102	-0.03904
Health & social work	1.364	-1.315	-0.02811	-0.1998
Professional & technical	-1.706	-1.281	-0.1485	-0.09176
Accommodation & food	0.9293	-1.16	-0.1014	-0.06816
Manufacturing	-1.726	-0.9774	0.002333	-0.06307
Other services	0.08662	-0.9615	-0.005363	-0.02747
Wholesale & retail	-0.2212	-0.9404	-0.00434	-0.09922
Transport & storage	-0.6101	-0.9088	0.01466	-0.0384
Administrative & support	-0.35	-0.8257	0.01769	-0.03341
Public admin & defence	1.532	-0.5018	0.08306	-0.04856
Construction	-0.4983	-0.3656	0.04711	-0.02206
Information & communication	-1.414	-0.3334	-0.1446	-0.01231
Real estate	-0.1885	-0.1449	-0.01189	-0.001545
Finance & insurance	0.173	-0.08175	0.02152	-0.003637
Education	-0.5587	0.7672	0.02436	0.07808
Agriculture, forestry & fishing	0.5335	0.8904	-0.06986	0.0122
Mining & quarrying	1.732	0.9175	0.05923	0.01771
Energy supply	0.6671	1.145	0.02925	0.01348

Notes: The industry names are abbreviated from the official SIC 2007 section titles.

C Scoring Protocol and Audit Trail

This appendix documents the frozen configuration in enough detail for the exercise to be reproduced or audited independently. Table A4 records it.

C.1 Frozen configuration

I fix a single model and freeze prompt before the main run. Decoding is deterministic to the extent the providers allow: the temperature is set to zero and no sampling parameter is varied across the run. To make any later change to the elicitation detectable, the scoring script records a SHA-256 hash covering the system message and the user-message template jointly, so the figures either reproduce from the recorded configuration or else fail to match the hash.

Table A4: Frozen scoring configuration

Item	Value
Baseline run	
Provider / model snapshot	Anthropic, `claude-sonnet-4-6`
Transport	Anthropic Batch API
Run date	8 July 2026
Temperature	0
Maximum output tokens	150
Top-p / top-k	not set (provider default)
Seed	none; the Messages API exposes no seed parameter
Retries / concurrency	2 retries; 8 concurrent requests on the synchronous path
Response format	single JSON object, keys `substitution`, `complementarity`, `primary_mode`, `key_factors`
Prompt SHA-256	8e831d4ea7c85bb9962c71a7fbf15672bb681b4fb33a61a1df47 35c8c6ea175a
V0 batch identifier	msgbatch_01Gvienm9x6cVRoVcpEcP7Qf
Cross-model arm (Section 5.5)	
Alternative provider	OpenAI, `gpt-4o-2024-11-20`, Batch API; `response_format = json_object`, same temperature and token cap
Alternative tier	Anthropic, `claude-haiku-4-5-20251001`, same transport as baseline
Elicitation	
Horizon	10 years, from a base year of 2026
Scoring scale	two integer scores on $[0, 100]$: substitution and complementarity
Criteria	six fixed considerations, presented in a fixed order

C.2 Prompt and rubric

The system message casts the model as an economist assessing what artificial intelligence can do to a stated O*NET task within the horizon, instructs it to assume capability progress continues at approximately the pace observed between 2020 and 2025, and asks for two scores: a substitution score for the extent to which AI could perform the task end-to-end with no meaningful human involvement, and a complementarity score for the extent to which it could raise the productivity of a worker who is retained. The six considerations the model is directed to weigh cover information-processing content, physical dexterity in unstructured settings, empathy and non-delegable accountability, genuine novelty, verifiability of output, and routinisation. The user message supplies the occupation title and the task text verbatim from O*NET. The full prompt text, the paraphrased variant and the remaining calibration perturbations are reproduced in the replication archive and can be provided on request.

D Calibration and Robustness Detail

D.1 Simulation sampling

The calibration draws $N = 90$ occupations, eighteen from each of five broad occupational families. The families are obtained by collapsing the O*NET two-digit major groups onto their leading digit, which yields a professional and technical family, a second covering community, legal, education, arts and healthcare practitioner work, a third covering healthcare support, protective, food, building and personal services, a fourth covering sales, office administration, farming, construction and installation, and a fifth covering production and transport. The partition is coarse deliberately. Because $\hat{\sigma}_g^2$ is estimated from four scoring conditions on the occupations drawn within each family, a finer split leaves the families too thin for the variance to be usable, and this is the finest grouping at which every family carries eighteen sampled occupations. The same $g(j)$ indexes the noise variance throughout Section 4.3 and the propagation of Table 4.

Within each family, 10 occupations are drawn at random and 8 are the ones sitting nearest the classification boundary. Estimating the framing variance on a purely boundary-proximate sample would bias it upward if ambiguity rises towards the middle of the scale, which is where the boundaries sit. The random stratum avoids this and supplies $\hat{\sigma}_g^2$; the boundary stratum is held out, and is used to check that instability concentrates where the construction says it should rather than to estimate anything.

D.2 Correlated-error propagation

Across $R = 1{,}000$ draws I perturb each occupation score, $s_j^{(r)} \sim N(s_j, \hat{\sigma}_{g(j)}^2)$ truncated to $[0, 100]$, re-aggregate, re-weight, and record $\Delta \bar{E}^{(r)}$. Because correlated framing errors would otherwise lead independent draws to understate aggregate uncertainty, I allow intra-family correlation $\varepsilon_g^{(r)} \sim N\Big(0, \hat{\sigma}_g^2[(1-\rho)I + \rho\mathbf{1}\mathbf{1}^\top]\Big)$ for $\rho \in \{0, 0.2, 0.4\}$. The resulting inflation is exactly the design-effect factor derived at Equation (24), which reduces to the independent case at $\rho = 0$; since ρ is not identified from four scoring conditions I do not estimate it, and read the spread across the ρ rows of Table 4 as the sensitivity of the reported uncertainty to the assumption.

E Additional Results

The compositional gap as a regression coefficient

The shift-share reading of Equation (3) can be restated as a regression, which shows that the differential survives within broad occupational families and not only across them.

Stacking occupation-by-region observations,

$$y_{k,r} = \alpha + \beta \bar{E}_k + \zeta\Big(\bar{E}_k \times \mathrm{Scot}_r\Big) + \phi_{s(k)} + u_{k,r}, \tag{25}$$

weighted by occupation-group employment, where $y_{k,r}$ is log occupational employment and, in a second specification, the employment share; $\phi_{s(k)}$ are one-digit SOC major-group fixed effects; and errors are clustered by occupation. Since $\bar{E}_k$ does not vary by region, ζ is identified purely from the interaction of the common exposure score with the regional employment distribution, and a negative estimate means composition insulates Scotland. With major-group fixed effects, $\widehat{\zeta}$ is -0.629 (SE 0.241) for log employment and -0.018 (0.007) for the employment share. Within broad occupation groups, Scottish employment sits systematically further down the exposure gradient than rUK employment.

Measurement-error benchmark for the wage gradient

Exposure enters the wage regression at the SOC minor-group level, so every worker in a minor group carries the same score and the measurement error is grouped rather than independent across observations. The classical correction $\widehat{\beta}_1^{\mathrm{EIV}} = \widehat{\beta}_1^{\mathrm{OLS}}/(1-\widehat{\lambda})$, with reliability complement $\lambda = \sigma_u^2/\operatorname{Var}(E_j)$ and $\sigma_u^2 = \widehat{\sigma}^2_{g(j)}/10^4$, assumes independent homoskedastic error per observation, and the violation runs in the direction that overstates the correction. By the Frisch–Waugh–Lovell theorem the effective reliability is computed on the variation in E left after residualising on the industry and year fixed effects and the worker controls of Equation (10), so any control that explains part of E worsens the attenuation. I report the classical figure as an upper benchmark on the attenuation rather than as the correction itself.

The formula is also a single-regressor result, which is why it cannot be read off term by term for the interaction. It applies once β_3 is read as the gap between the rUK exposure slope, β_1, and the Scottish one, $\beta_1+\beta_3$. Each is a single mismeasured regressor within its own region, and since σ_u^2 is a property of the shared occupation score while $\operatorname{Var}(E)$ differs across regions only through employment composition, the two reliabilities are equal to a close approximation and the common factor $1/(1-\widehat{\lambda})$ carries the interaction as well.

The reliability complement is small, $\widehat{\lambda} = 0.016$, because the between-occupation variation in exposure dwarfs the scoring noise. The benchmark nudges the exposure slope up by 1.6%, from 1.276 to 1.297, and the interaction by the same factor, from -0.300 to -0.305. Neither departure noted above is large enough to matter at that reliability. Overall, the correction is immaterial even under an estimator whose violated assumptions overstate it, so the grouped structure of the measurement error cannot rescue an attenuation this small.

Table A5: Scoring measurement error in the wage gradient: OLS and the classical benchmark

	OLS	EIV benchmark	SE (OLS)
Exposure (β_E)	1.276	1.297	0.170
Exposure × Scotland ($\beta_{E\times\text{Scot}}$)	−0.300	−0.305	0.065

Notes: Worker-level regression of Equation (10) on the pooled 2022–2025 APS, N = 176,444, weighted by the APS income weight, with industry and year fixed effects and standard errors clustered by occupation, the level at which the exposure regressor varies; the occupation × region figure of 0.226 on the Scotland interaction is reported in Section 5.6 as a comparison. The EIV benchmark applies $1/(1-\widehat{\lambda})$ with $\widehat{\lambda} = 0.016$ to both the exposure slope and the interaction, the latter under the equal-reliability argument in the text.

F Specifications and Projection Detail

I derive the fiscal extension in full, along with the parameter it inherits.

F.1 Productivity calibration

Where Equation (11) comes from. Let the final good be produced under constant returns from a set of tasks with prices $p_1, \ldots, p_N$ and unit cost function $c(p_1, \ldots, p_N)$, homogeneous of degree one. By Shephard's lemma the cost-minimising task demands are $\partial c/\partial p_n$, so

$$\varsigma_n \;=\; \frac{\partial \ln c}{\partial \ln p_n} \;=\; \frac{p_n\,(\partial c/\partial p_n)}{c} \tag{26}$$

is the Shephard cost share of task n, and the shares sum to one by Euler's theorem under linear homogeneity. If AI lowers the cost of task n by the fraction κ_n, so that $d\ln p_n = -\kappa_n$, then totally differentiating gives $d\ln c = \sum_n \varsigma_n\, d\ln p_n = -\sum_n \varsigma_n \kappa_n$, and since total factor productivity is the inverse of unit cost at fixed input prices,

$$\Delta \ln \text{TFP} \;=\; -\,d\ln c \;=\; \sum_n \varsigma_n\, \kappa_n. \tag{27}$$

This is Hulten (1978) applied at the task rather than the sector level, which is the step Acemoglu (2025) takes within the task framework of Acemoglu and Restrepo (2019). Aggregating Equation (27) over a region's employment distribution requires one further step, and it is where the second parameter of Equation (11) enters. Set $\kappa_n = \kappa$ on tasks AI performs and zero otherwise. Exposure identifies the task content AI could perform, and only a fraction π of that content is profitably automated over the horizon, so the displaced cost share of minor group k is $\pi\bar{E}_k$ rather than $\bar{E}_k$. Substituting into Equation (27) and weighting by employment gives Equation (11).

The imported parameters. The average cost saving on an exposed task, κ, rests on the two task-level randomised experiments Acemoglu (2025) relies on: Noy and Zhang (2023) on mid-level professional writing, and Brynjolfsson et al. (2025) on customer-support agents. He reads these as labour cost savings of 40% and 14% respectively, and converts their average of 27% into an overall cost saving by multiplying through a labour share of 0.57, giving $0.27 \times 0.57 = 0.154$. That is the central setting, and it reproduces his headline: $0.199 \times 0.23 \times 0.154 = 0.0071$, the 0.71% ten-year TFP gain he reports as an upper bound.

The low setting comes from his refinement for hard-to-learn tasks. Splitting exposed tasks into easy and hard, he assigns the 27% saving to the easy 74% and a quarter of it to the remainder, giving overall savings of 0.154 and 0.040, so that $0.033 \times 0.154 + 0.012 \times 0.040 = 0.0056$ and the ten-year gain falls to 0.55%. Since κ is constant across occupations here, I carry that refinement as a combined saving over all impacted tasks, $0.0056/0.045 = 0.124$. Acemoglu regards the hard-task figure as the more reasonable of the two, so the low column is his preferred case rather than a pessimistic tail. The high setting is the Office for Budget Responsibility (2025) figure: a task-level cost saving of 32% against the same 0.57 labour share, giving $0.32 \times 0.57 = 0.182$. The OBR pairs it with an automatable fraction of 0.33 and an exposure share of 0.40, neither of which I adopt.

The automatable fraction is $\pi_k = \pi = 0.23$ for every occupation, the share of exposed tasks Acemoglu judges profitably automatable over the decade, taken from Svanberg et al.'s computer-vision estimates. Because π_k does not vary across k, Corollary 3 applies and the regional differential inherits the cancellation of Proposition 1 exactly, while the levels are proportional to π and carry its full model dependence. The reported levels (2.36% for rUK, 2.33% for Scotland) sit above Acemoglu's 0.71% national figure because my continuous index is broader than his roughly 20% task-exposure measure, which is a further reason why I place no weight on the levels.

F.2 The elasticity of NSND revenue to earnings

The third step of Section 6.2 needs the proportional change in Scottish NSND revenue produced by a permanent proportional change in earnings. I could locate no Scotland-specific figure, but since the object is a standard one, I follow the UK approach of Creedy and Gemmell (2004) to acquire the revenue elasticity of a progressive income tax.

If every taxpayer's income rises by the fraction δ and the thresholds do not move, taxpayer i's liability rises by $m_i y_i \delta$, where m_i is the marginal rate at y_i, so

$$\varepsilon = \frac{\sum_i \omega_i \, m_i \, y_i}{\sum_i \omega_i \, T_i} = \frac{\text{income-weighted average marginal rate}}{\text{average rate on total income}}, \tag{28}$$

with ω_i the APS income weight and T_i the liability under the prevailing Scottish schedule, the personal allowance taper carried explicitly. Progressivity alone puts $\varepsilon > 1$; the frozen higher, advanced and top rate thresholds are what keep it there as earnings grow (Scottish

Government 2026).

On my pooled APS data, I find the income-weighted average marginal rate is 0.296 against an average rate on total income of 0.154, so $\varepsilon = 1.922$. Progressivity very nearly doubles the revenue consequence of a given earnings differential.

That figure is high but not unusual. García-Miralles et al. (2026) compute the same elasticity by microsimulation for 21 European income tax systems and find values between 1.7 and 2.0 in fourteen of them; Price et al. (2015) report an OECD average of about 1.85. They also find that elasticities rise with the phase-out of allowances, and the Scottish personal allowance taper is exactly such a phase-out.

One caveat concerns coverage. The APS records gross pay for employees only, so it excludes self-employment and pension income such that the upper end of the tail is thinner, which is where marginal rates are highest. Grossed by the income weight, the pooled data implies a liability of roughly three-fifths of the HMRC outturn per wave (Table A6; the pooled ratio of 2.46 sets four stacked waves against a single year of outturn). A fatter upper tail would raise the numerator and the denominator of Equation (28) together, but the marginal rate is bounded above by the top rate while the average rate is not, so the denominator would rise by proportionally more. I therefore read 1.922 as the top of a narrow range.

Table A6: Elasticity of Scottish NSND revenue to earnings

Quantity	Value
Income-weighted average marginal rate	0.296
Average rate on total income	0.154
Elasticity of NSND revenue to earnings, ϵ	1.922
Incidence-weighted elasticity, ϵ_E	1.974
Implied aggregate liability, pooled 4 waves (GBP m)	45786
Implied aggregate liability, per wave (GBP m)	11447
HMRC outturn, 2024-25 (GBP m)	18635
Coverage ratio, pooled (implied / outturn)	2.46
Coverage ratio, per wave (implied / outturn)	0.61
Scottish observations	18,550
Tax year of schedule	2026-27